

```
!pip install ultralytics
```



Collecting ultralytics

Downloading ultralytics-8.3.58-py3-none-any.whl.metadata (35 kB)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: tqdm>=4.64.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: pandas>=1.1.4 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Requirement already satisfied: seaborn>=0.11.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Collecting ultralytics-thop>=2.0.0 (from ultralytics)

Downloading ultralytics_thop-2.0.13-py3-none-any.whl.metadata (9.4 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from matplotlib)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib)

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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas)

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Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from torch)

Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.10/dist-packages (from torch)

Requirement already satisfied: networkx in /usr/local/lib/python3.10/dist-packages (from torch)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (from torch)

Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from torch)

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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.10/dist-packages (from sympy)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from Jinja2)

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Downloading ultralytics_thop-2.0.13-py3-none-any.whl (26 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.3.58 ultralytics-thop-2.0.13

```
import os
```

```
train_path = "/content/drive/MyDrive/License plate detection and recognition/train/images"
```

```
val_path = "/content/drive/MyDrive/License plate detection and recognition/valid/images"
```

```
print("Train path exists:", os.path.exists(train_path))
```

```
print("Validation path exists:", os.path.exists(val_path))
```



Train path exists: True

Validation path exists: True

```
import os
```

```
base_path = "/content/drive/MyDrive/License plate detection and recognition"
```

```
paths = [  
    os.path.join(base_path, "train/images"),  
    os.path.join(base_path, "valid/images"),  
    os.path.join(base_path, "test/images")  
]
```

```
for path in paths:  
    print(f"Path: {path}, Exists: {os.path.exists(path)}")
```

```
➞ Path: /content/drive/MyDrive/License plate detection and recognition/train/images, Exists: True  
    Path: /content/drive/MyDrive/License plate detection and recognition/valid/images, Exists: True  
    Path: /content/drive/MyDrive/License plate detection and recognition/test/images, Exists: True
```

```
from ultralytics import YOLO
```

```
# Load the YOLOv8 model
```

```
model = YOLO("yolov8n.pt") # Use 'yolov8n.pt' for Nano, or switch to 'yolov8s.pt' for Small, etc.
```

```
# Train the model
```

```
model.train(  
    data="/content/drive/MyDrive/License plate detection and recognition/data.yaml", # Path to the dataset  
    epochs=5, # Number of epochs  
    imgsz=640, # Image size  
    batch=16 # Batch size (adjust based on GPU memory)  
)
```



```

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```

```

import os
source = '/content/drive/MyDrive/License plate detection and recognition/test/images'
if os.path.exists(source):
    print("Path exists")
else:
    print("Path does not exist")

```

⇒ Path exists

```

from ultralytics import YOLO

# Load the trained model
model = YOLO('runs/detect/train/weights/best.pt')

# Run inference on the test dataset
results = model.predict(source='/content/drive/MyDrive/License plate detection and recognition/test/images',

# Display results
from IPython.display import Image, display
import glob

# Show saved images with predictions
image_files = glob.glob('runs/detect/predict/*.jpg') # Path where YOLO saves predictions
for image_file in image_files:
    display(Image(filename=image_file))

```



image 1/200 /content/drive/MyDrive/License plate detection and recognition/test/images/00bc6c35-3eec-4f
image 2/200 /content/drive/MyDrive/License plate detection and recognition/test/images/011d8b71-8bb9-4c
image 3/200 /content/drive/MyDrive/License plate detection and recognition/test/images/04977a14-f268-4d
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[illegible]

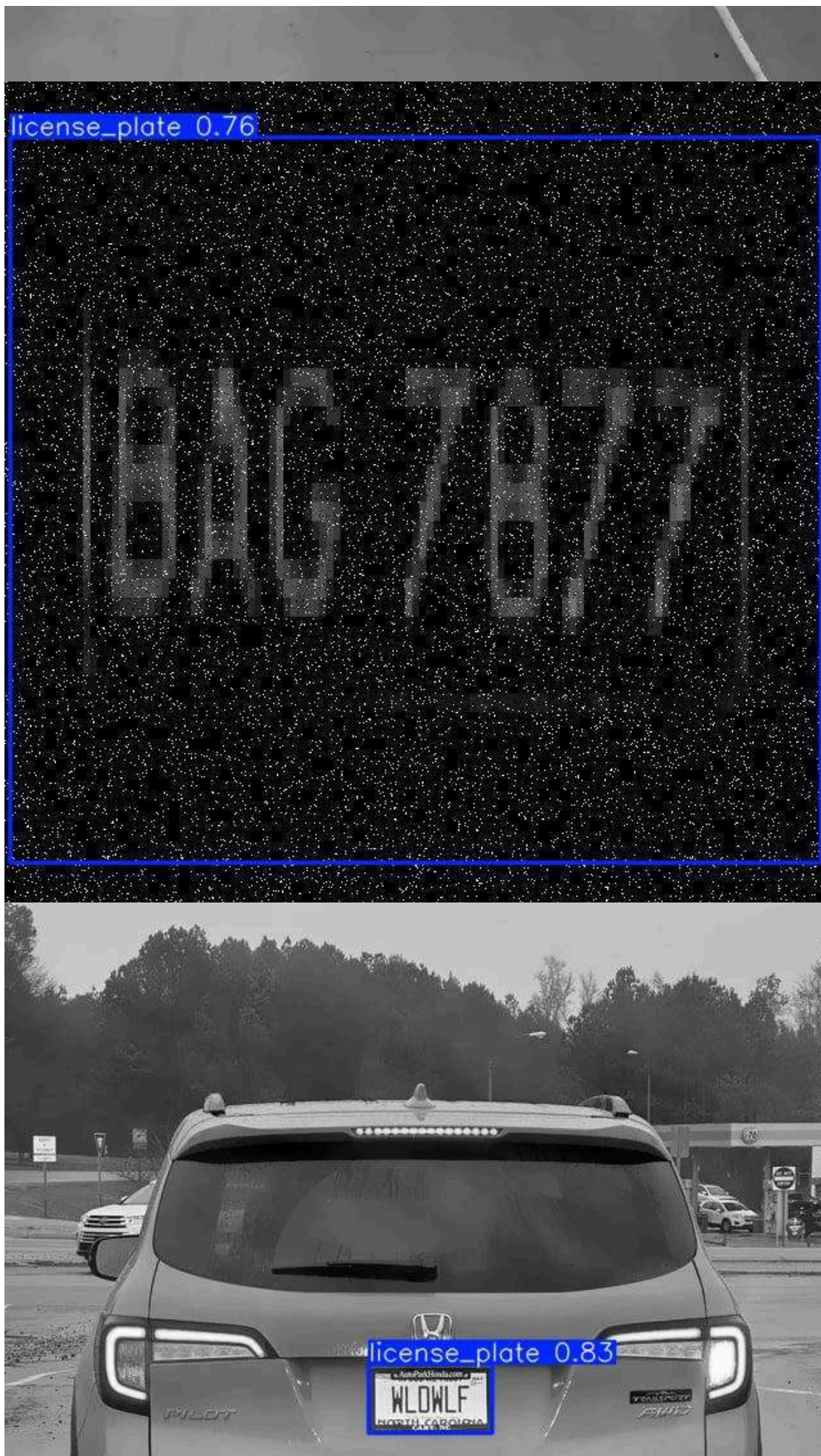
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Results saved to runs/detect/predict

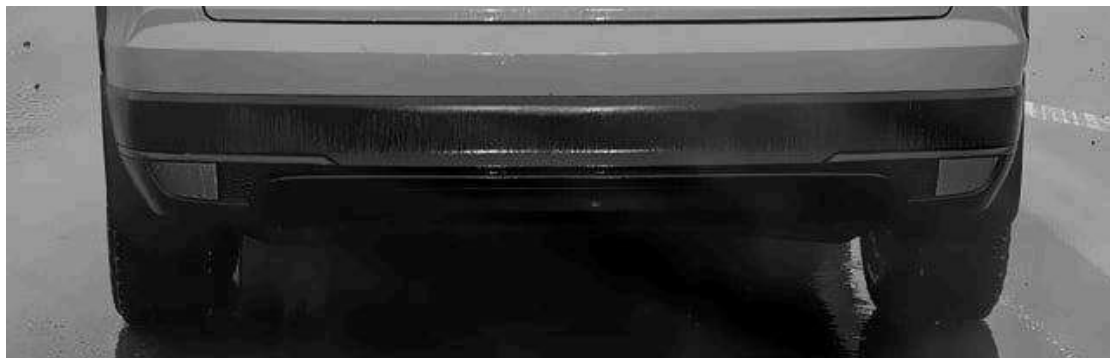




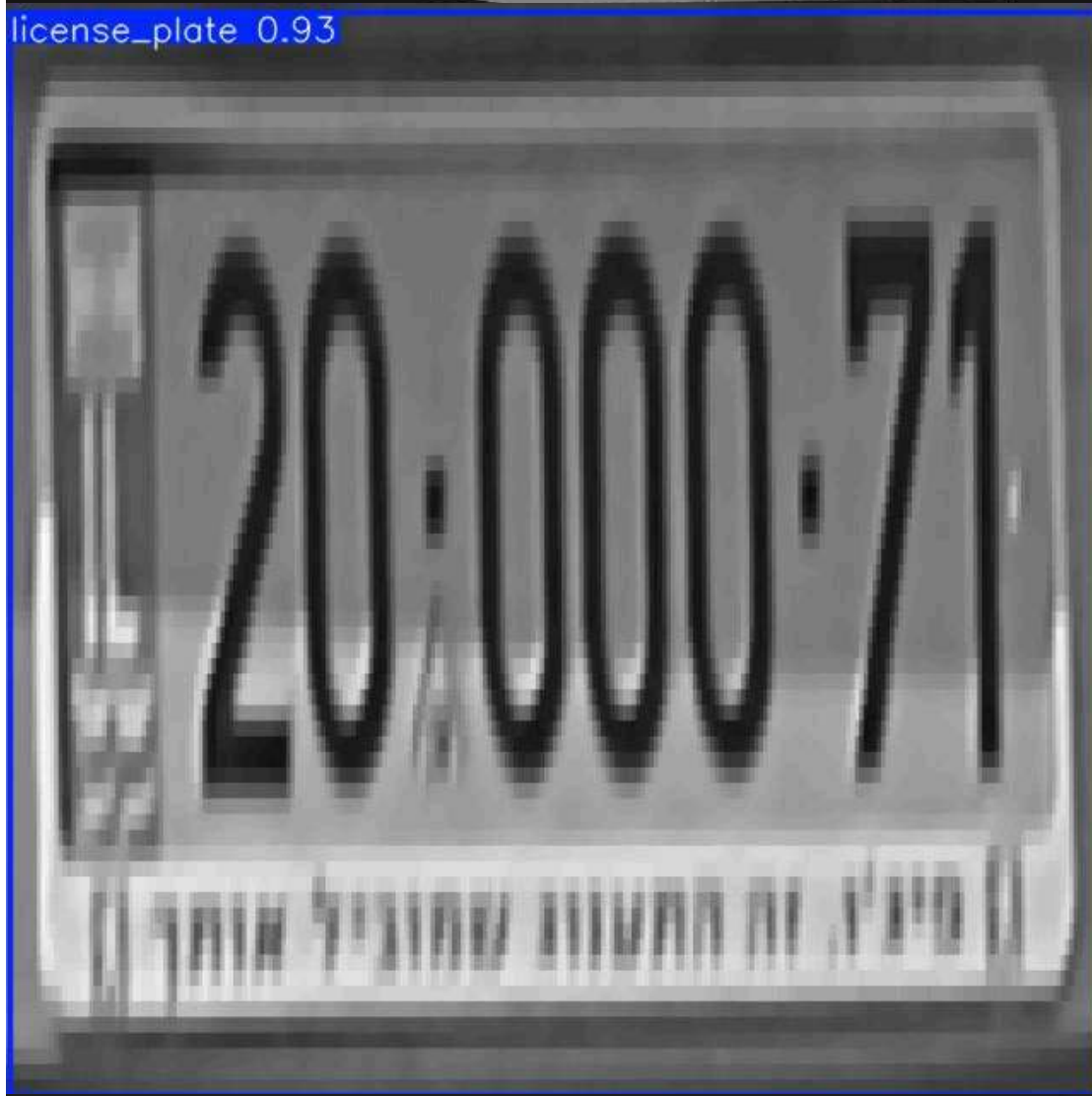




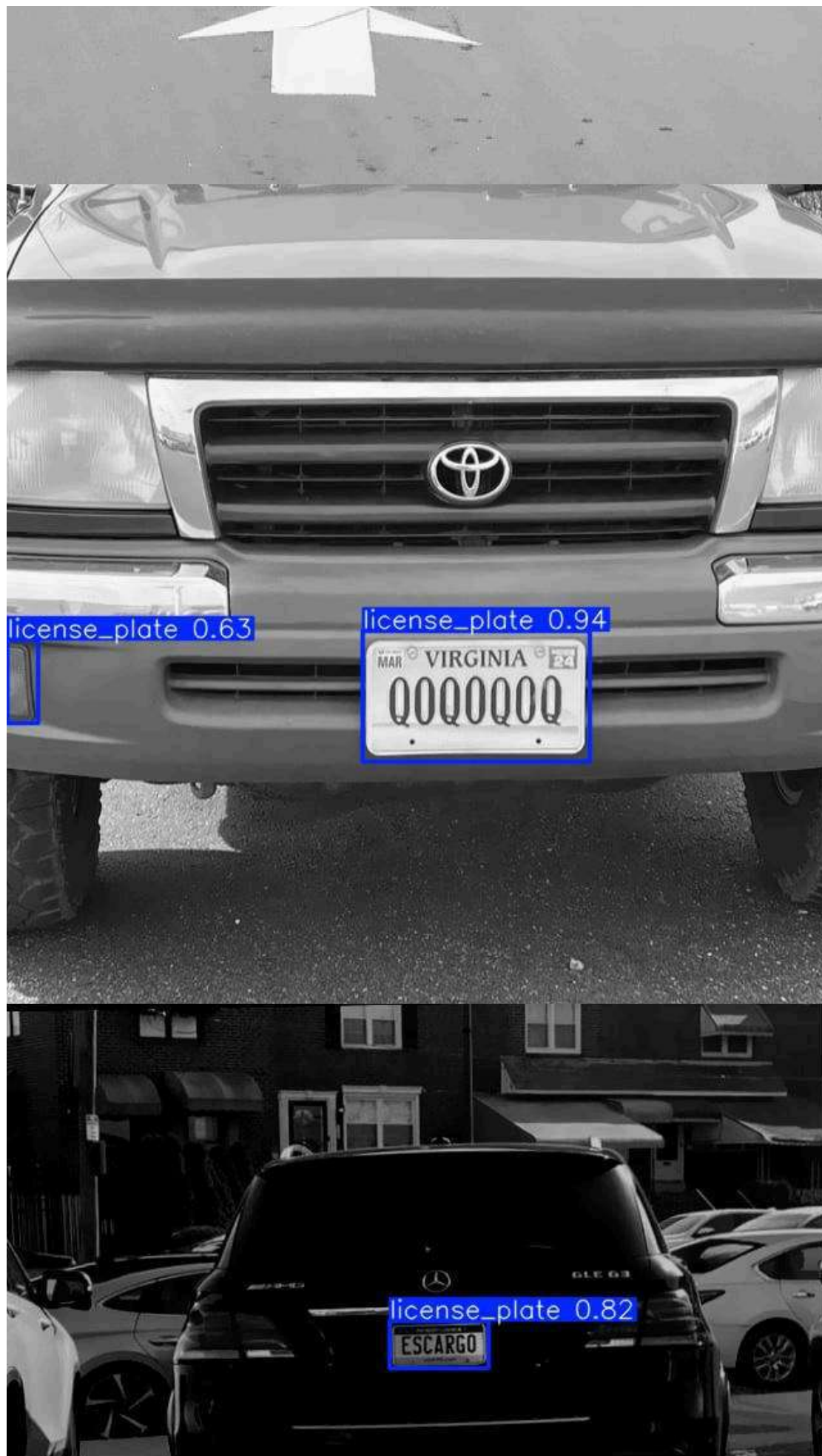


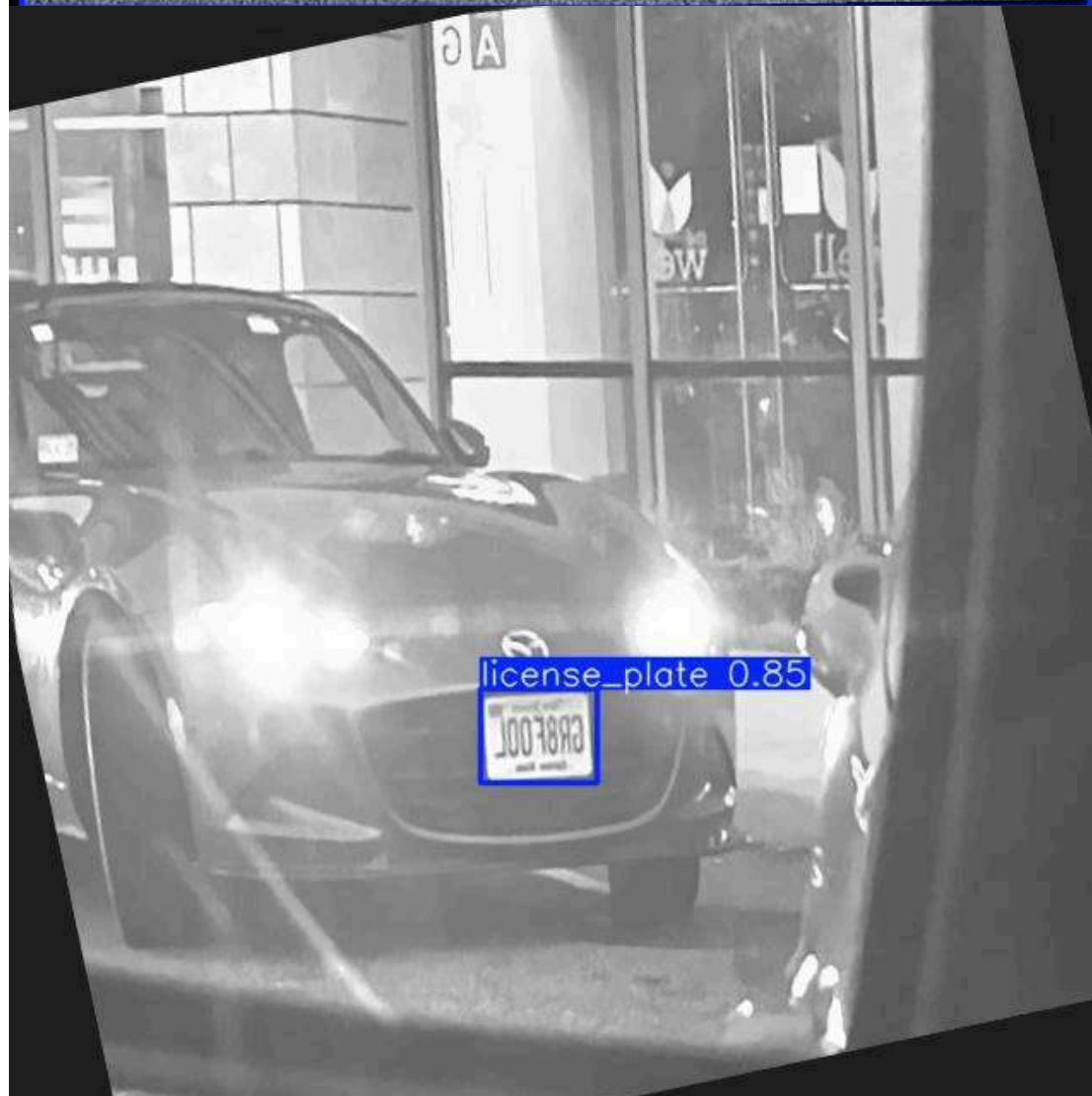
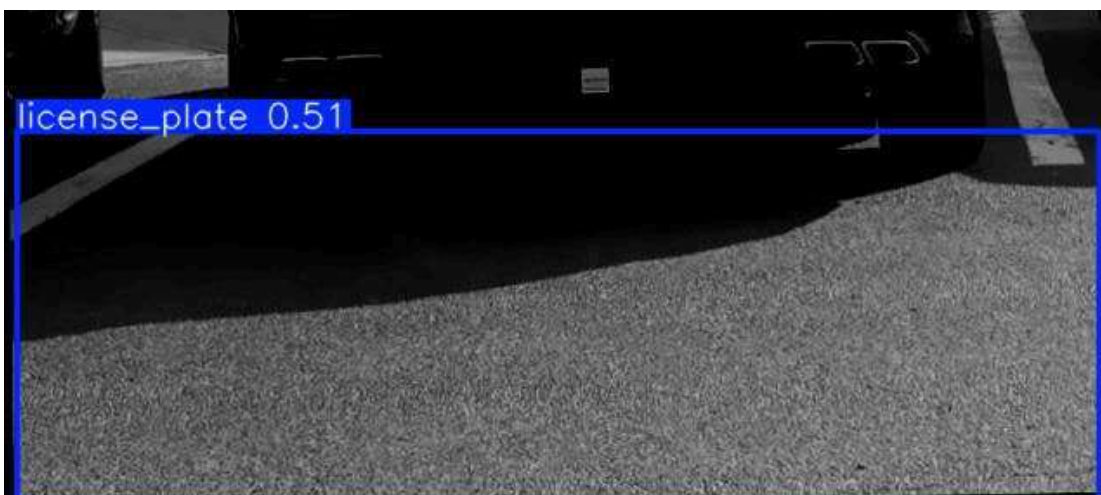










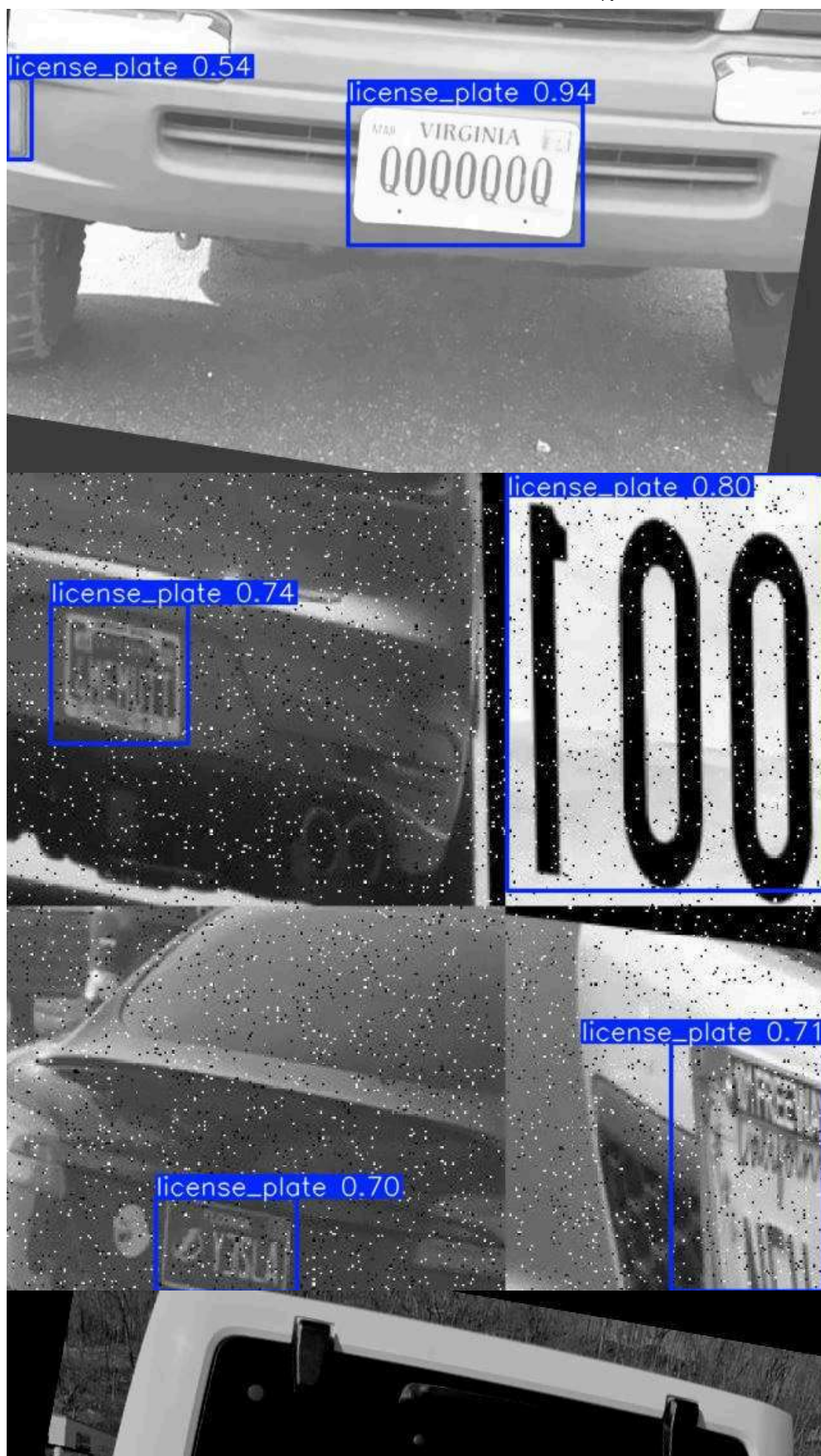


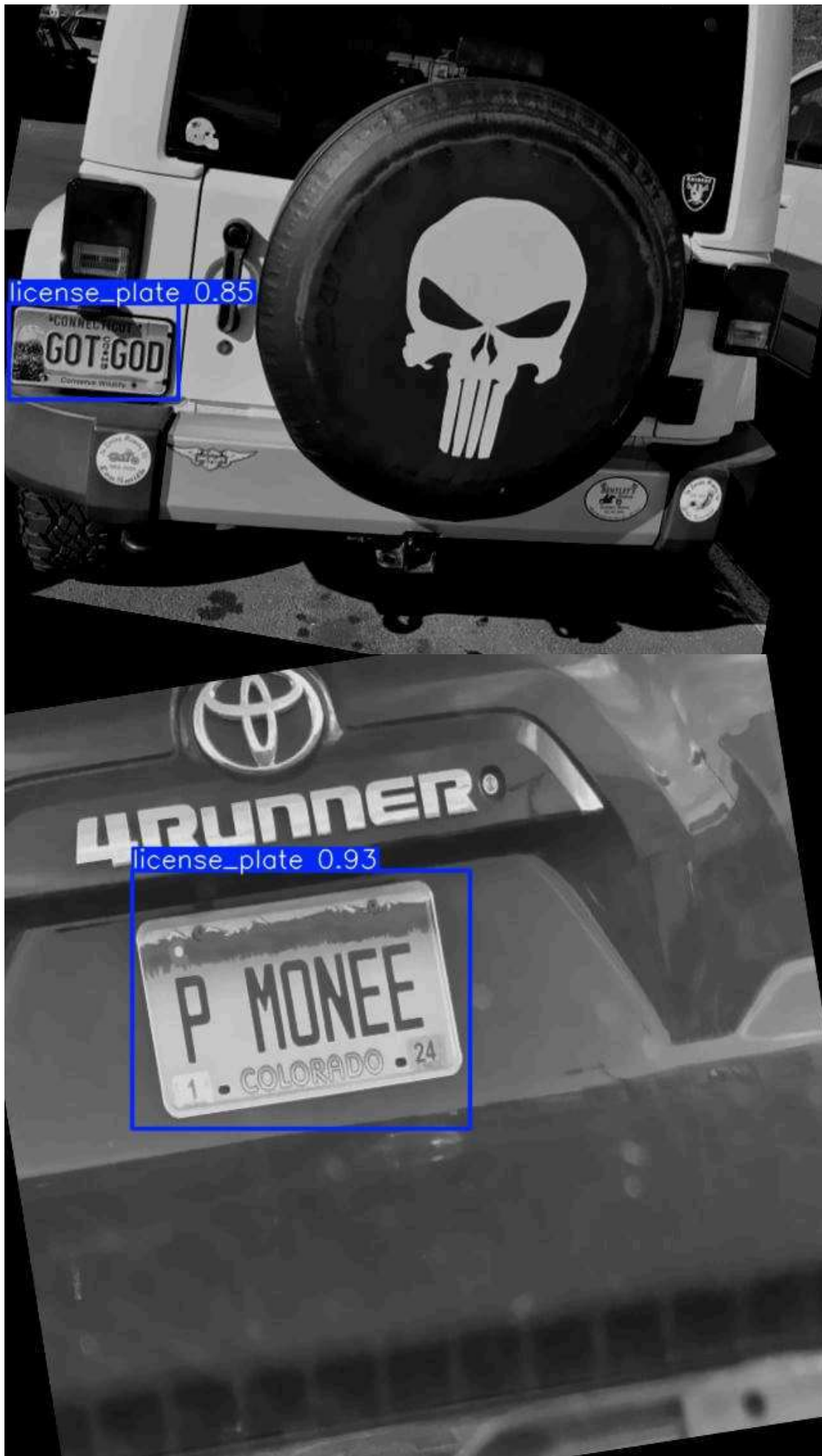












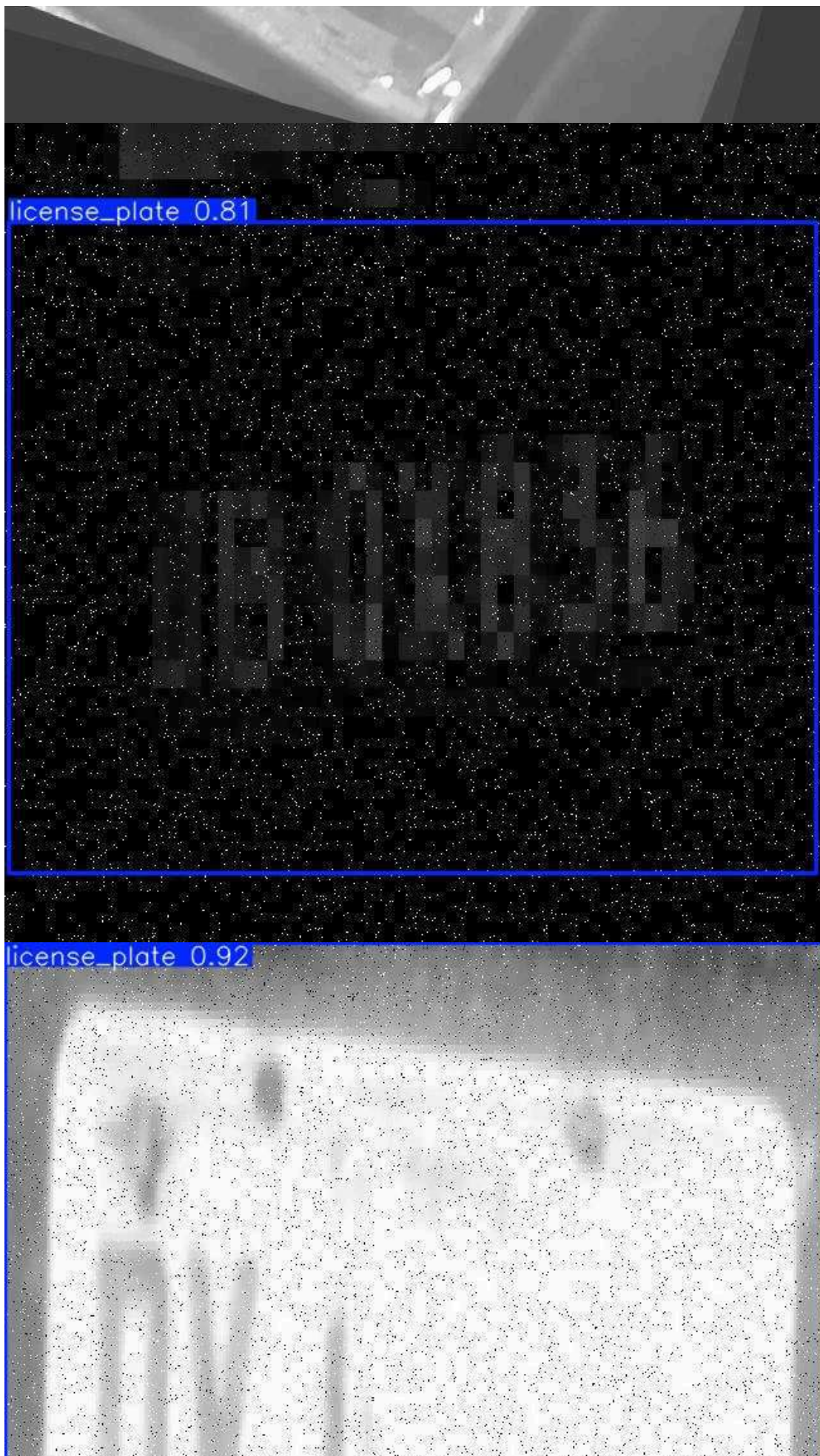


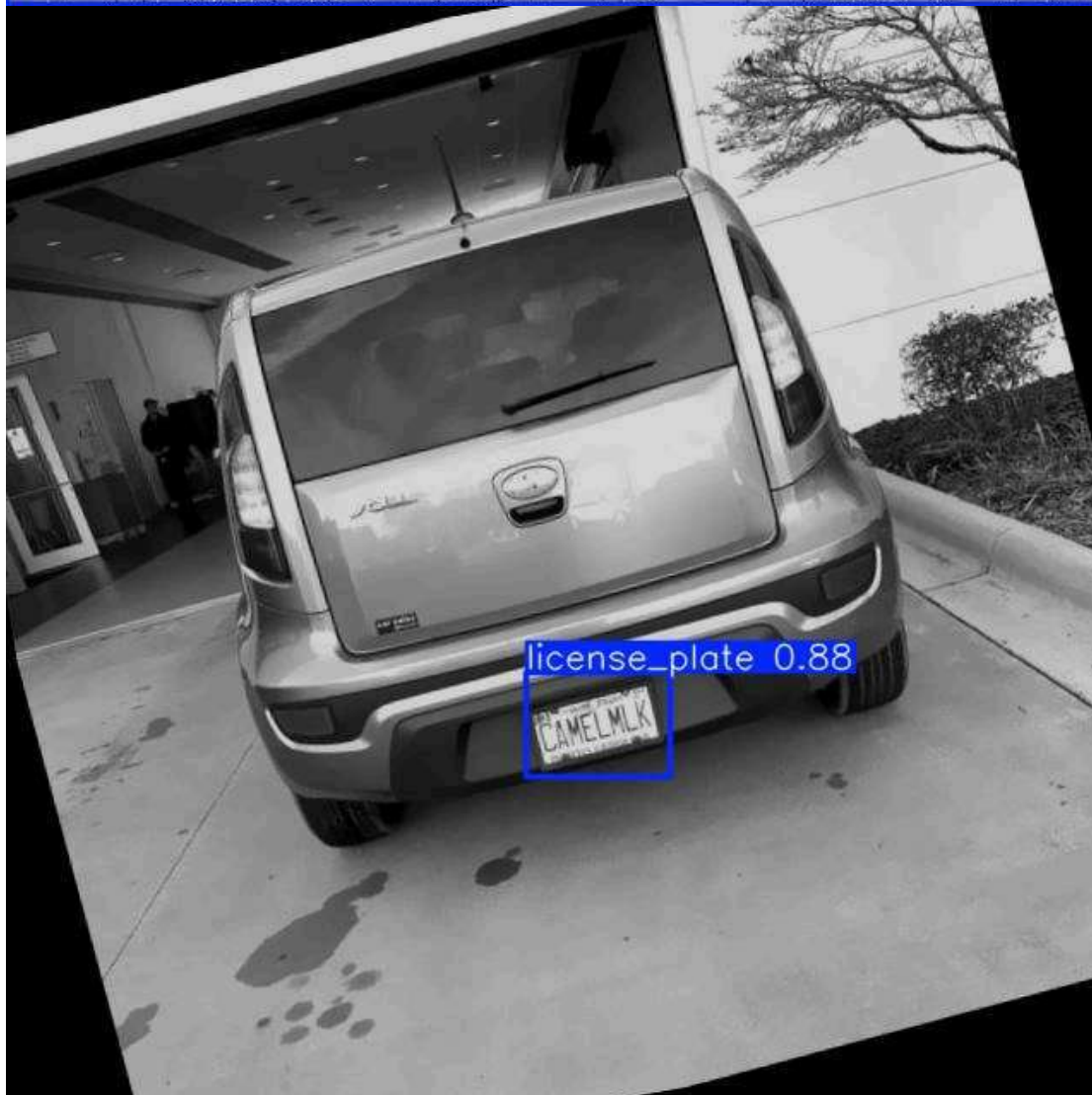
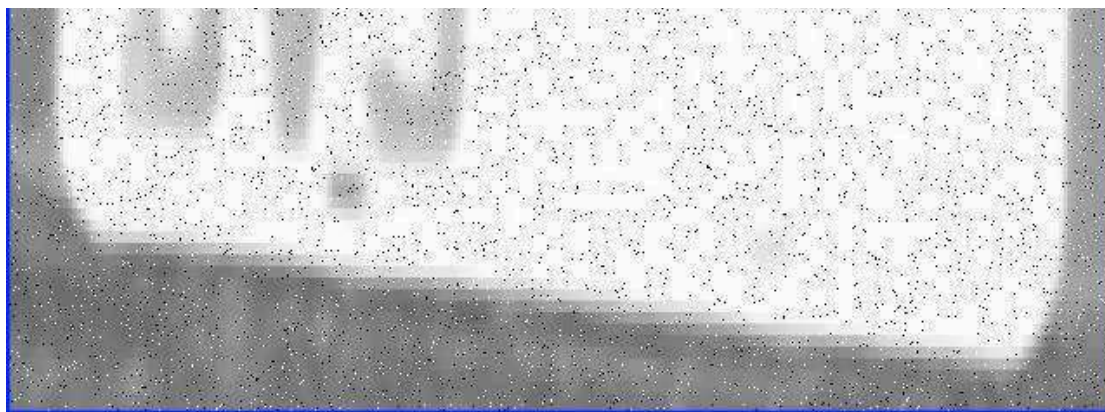


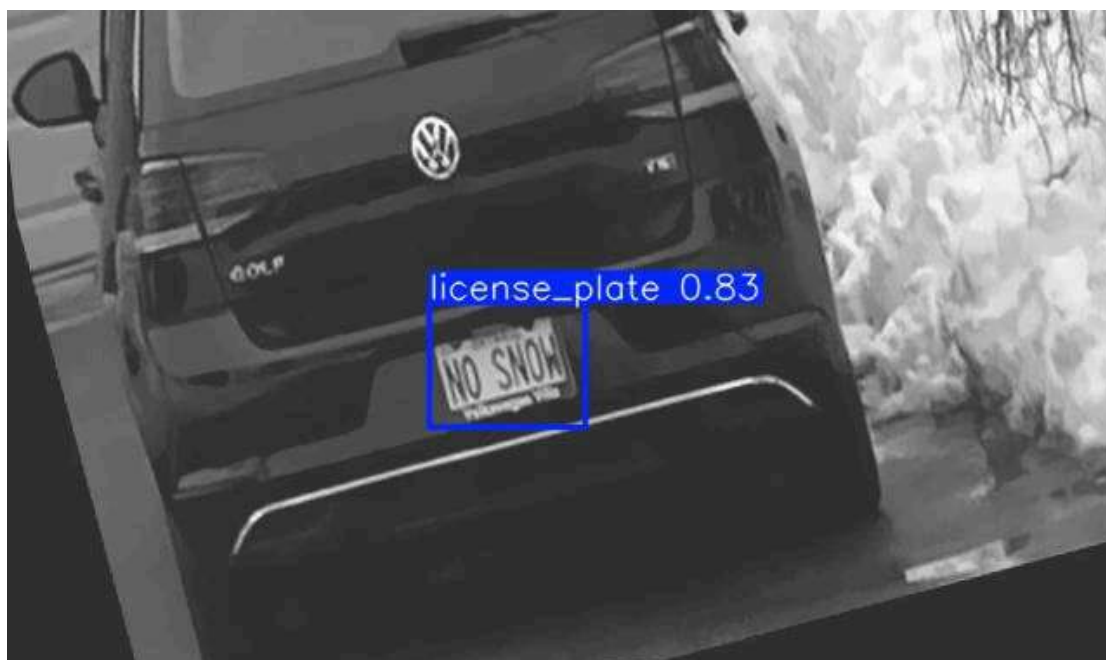








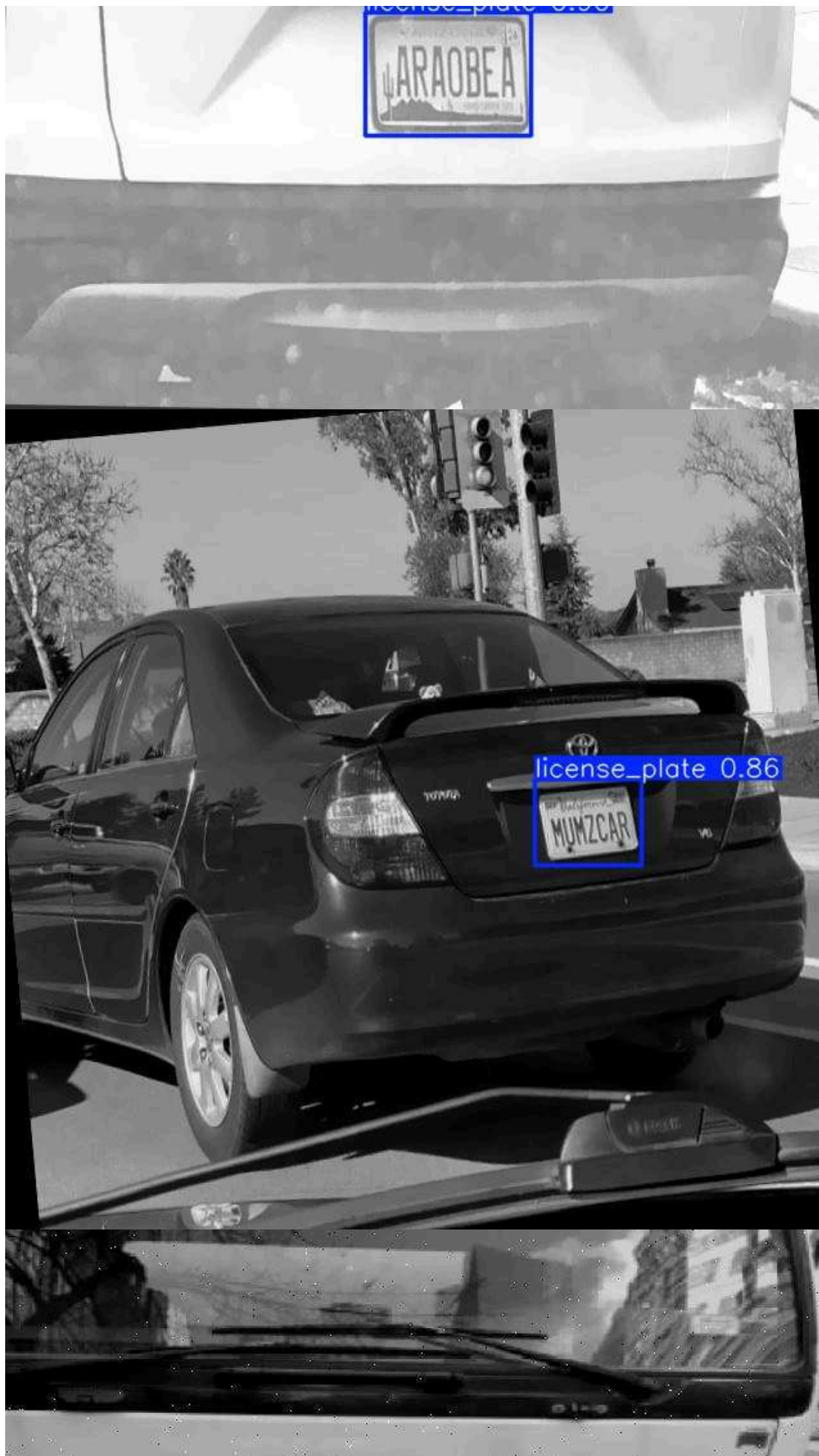














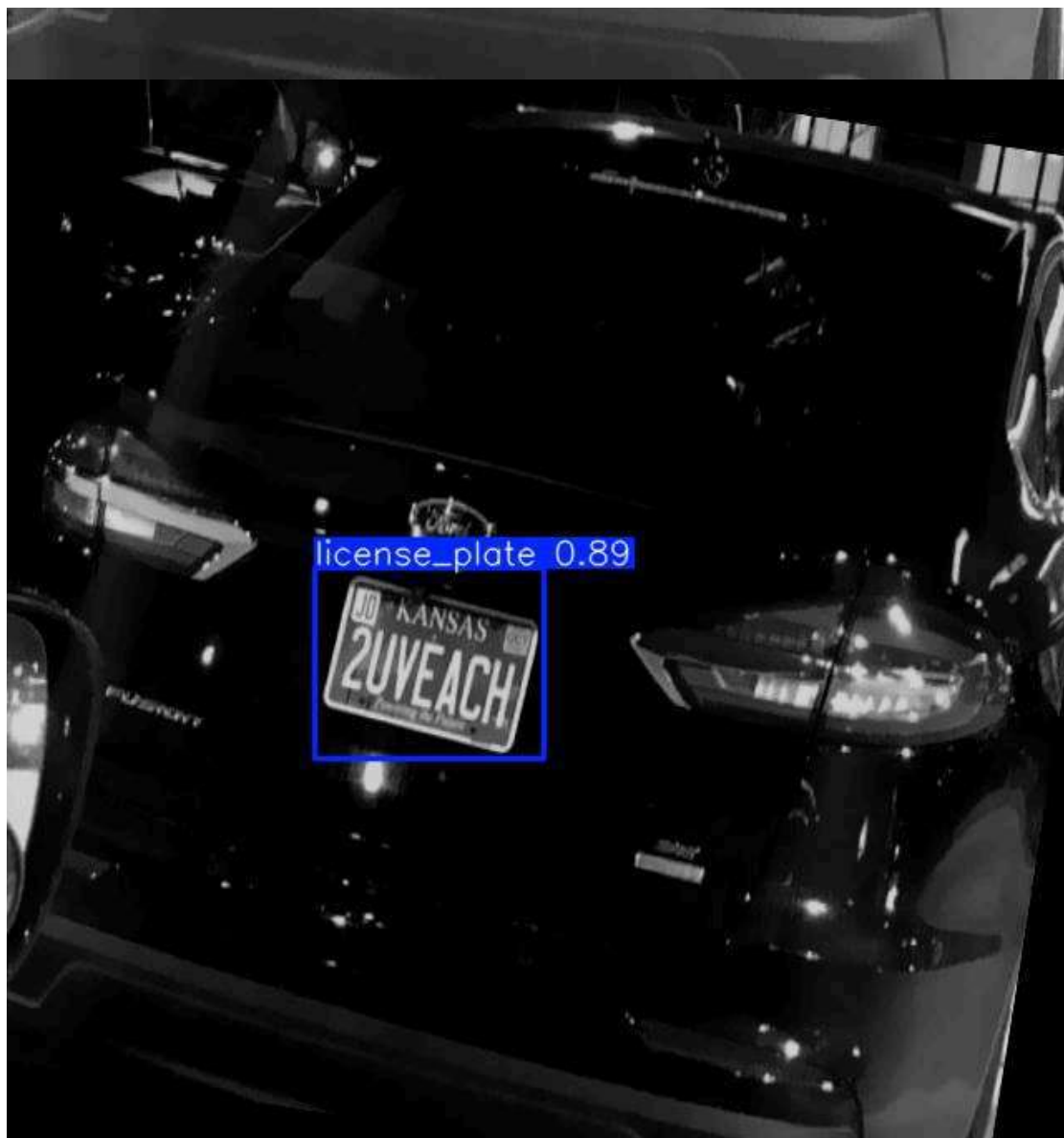








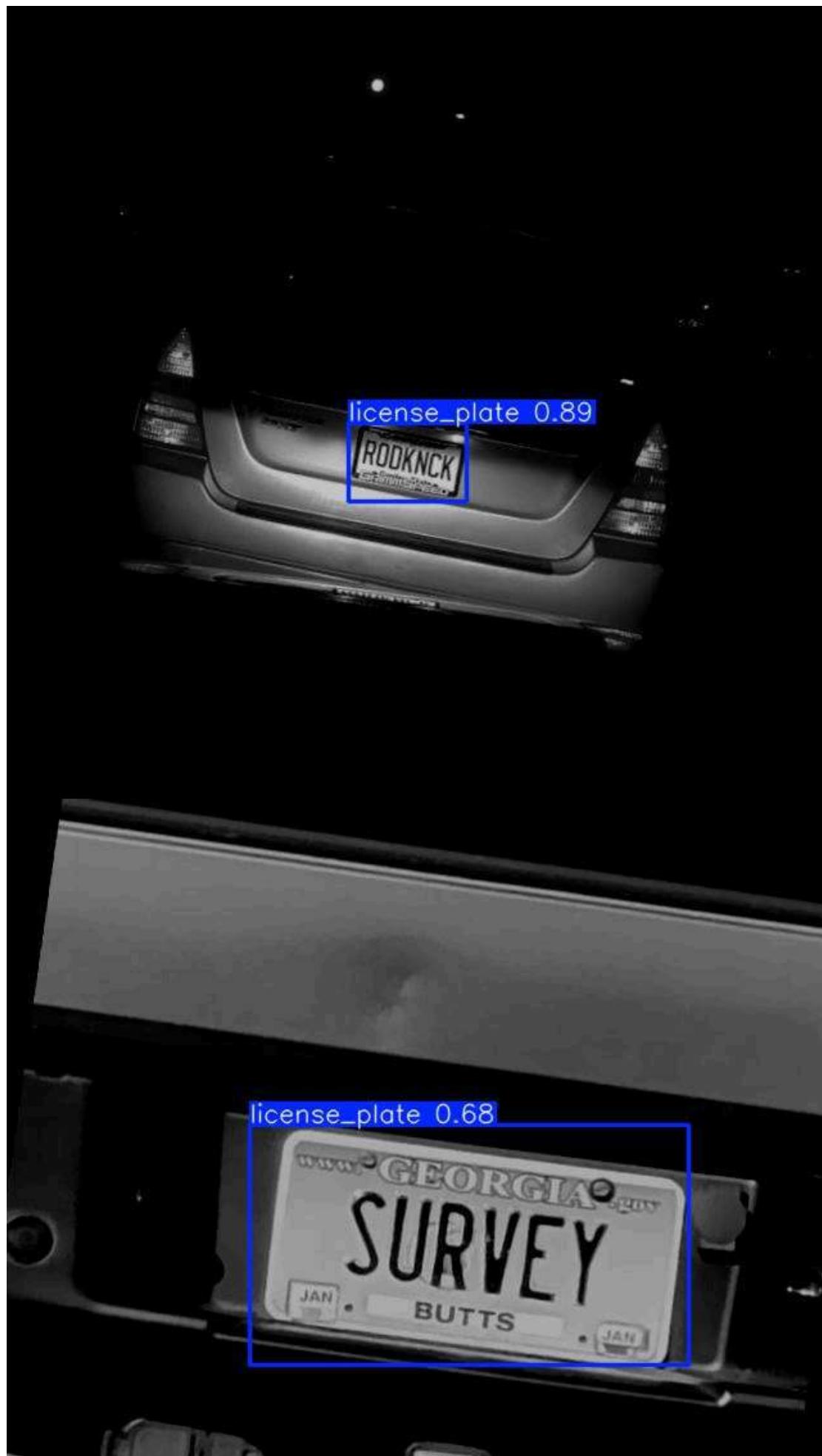


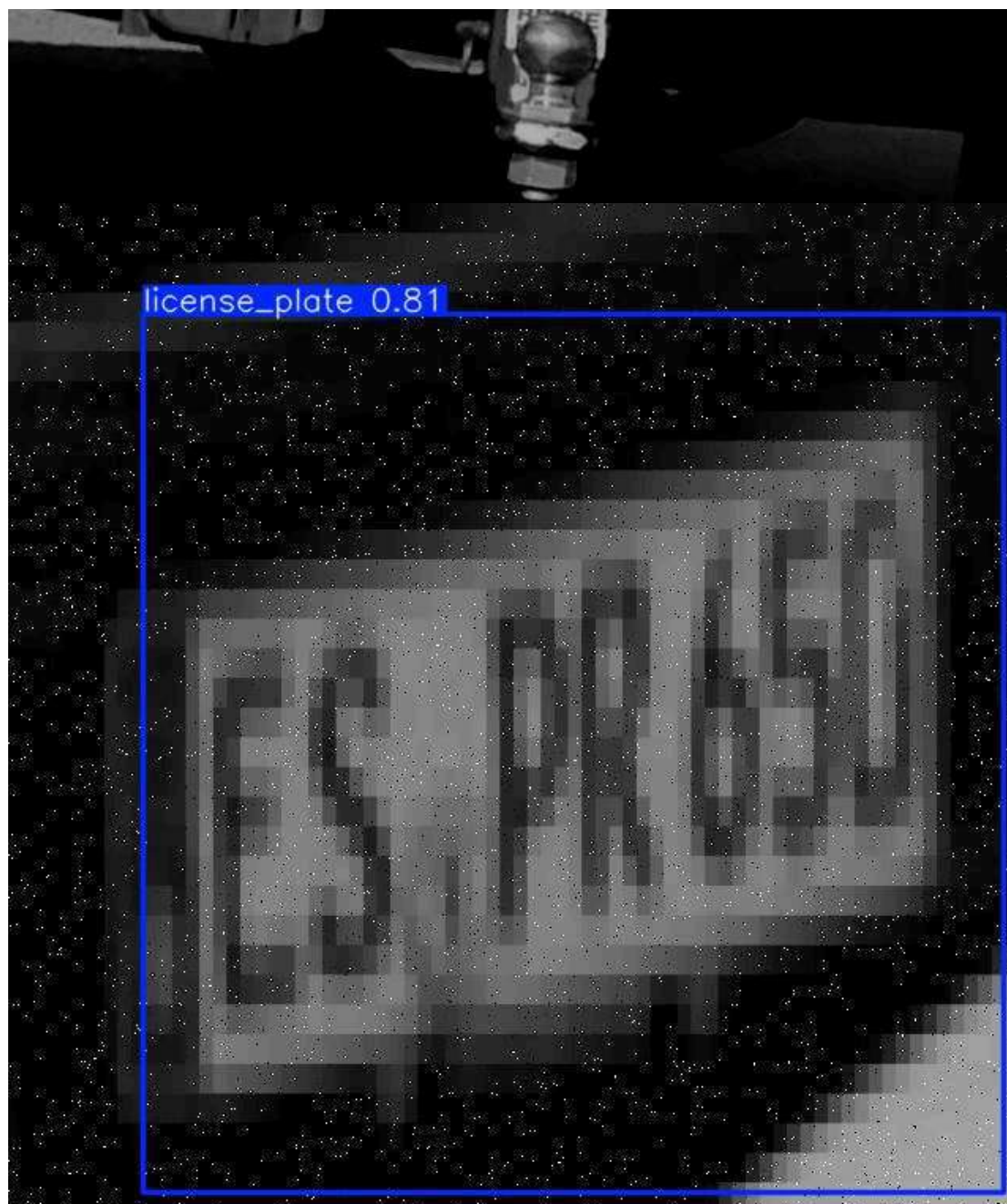












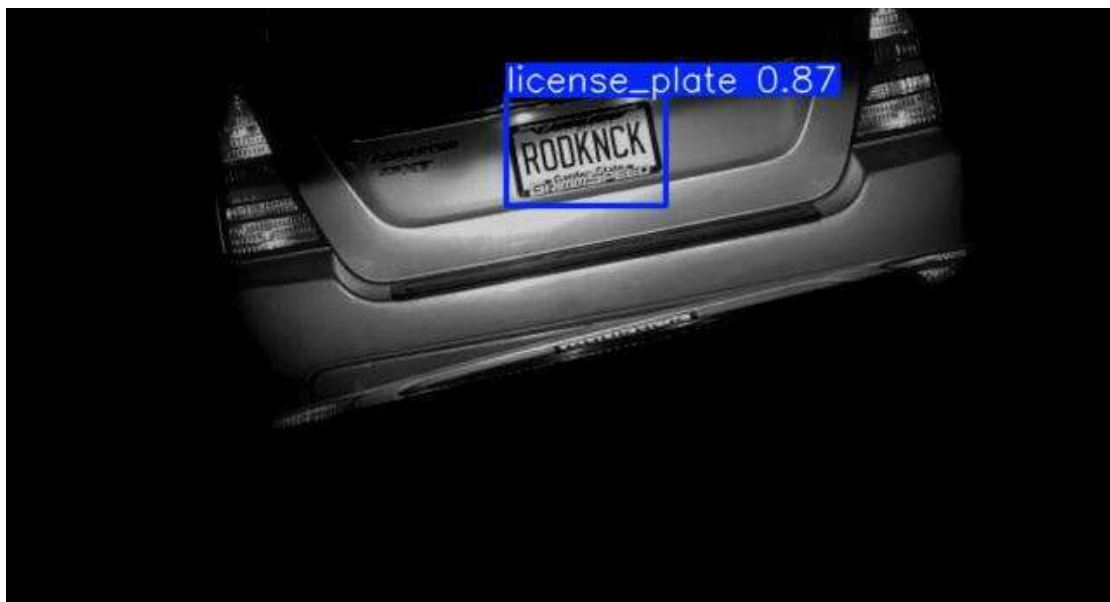










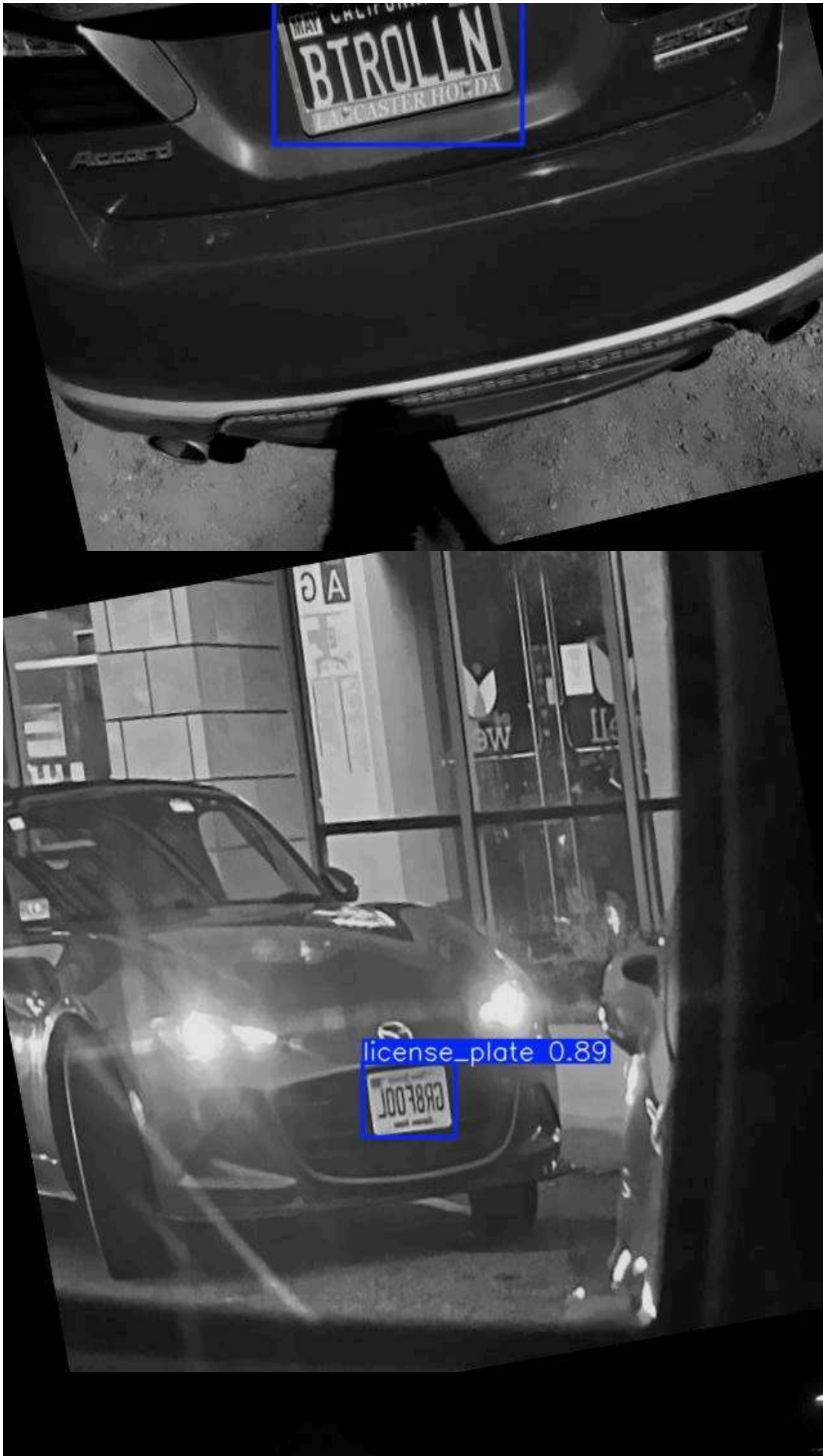






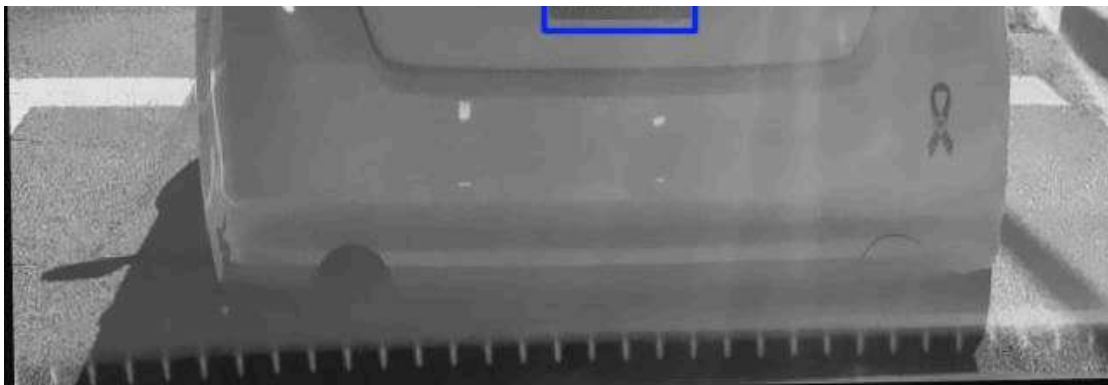














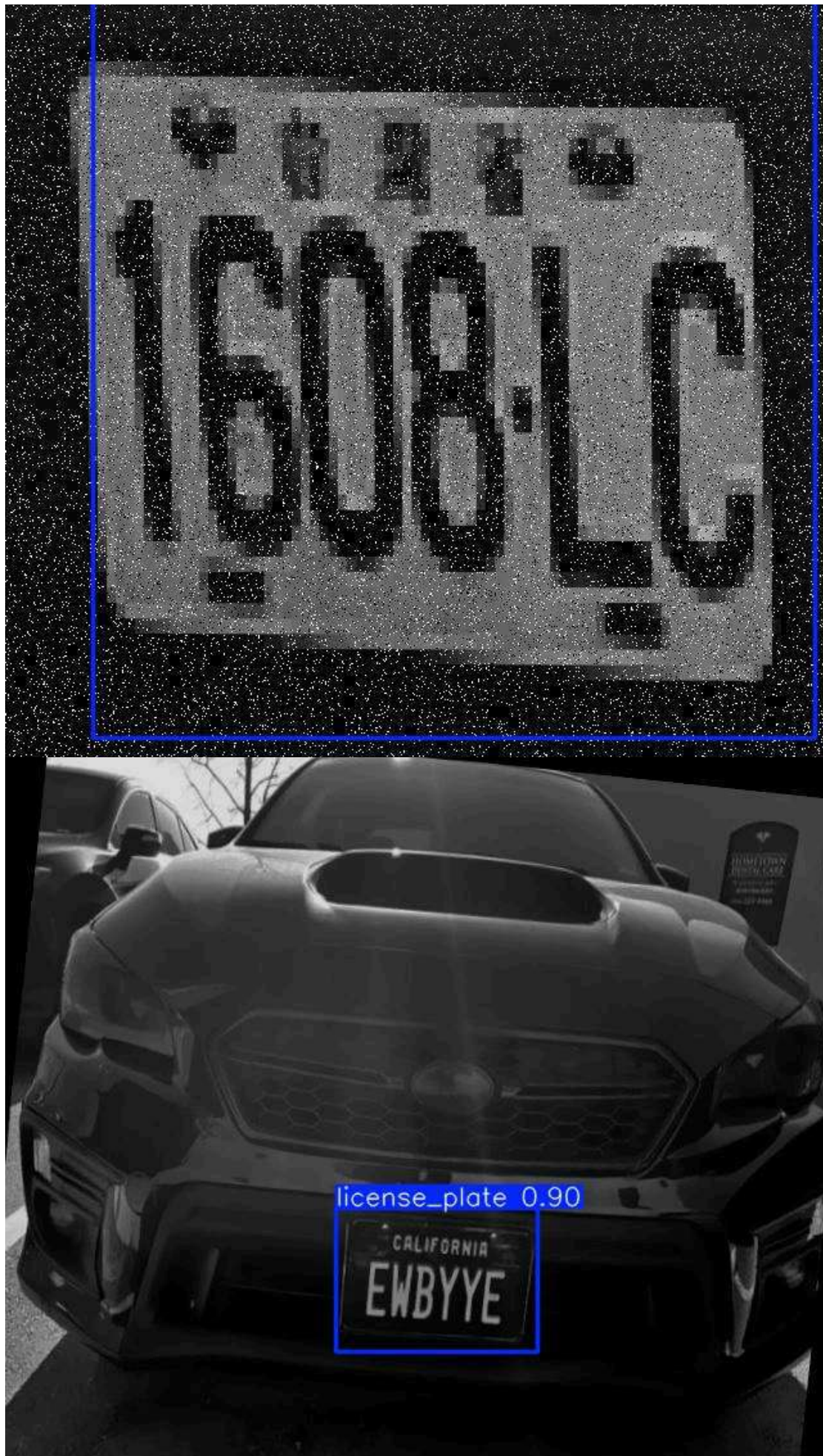






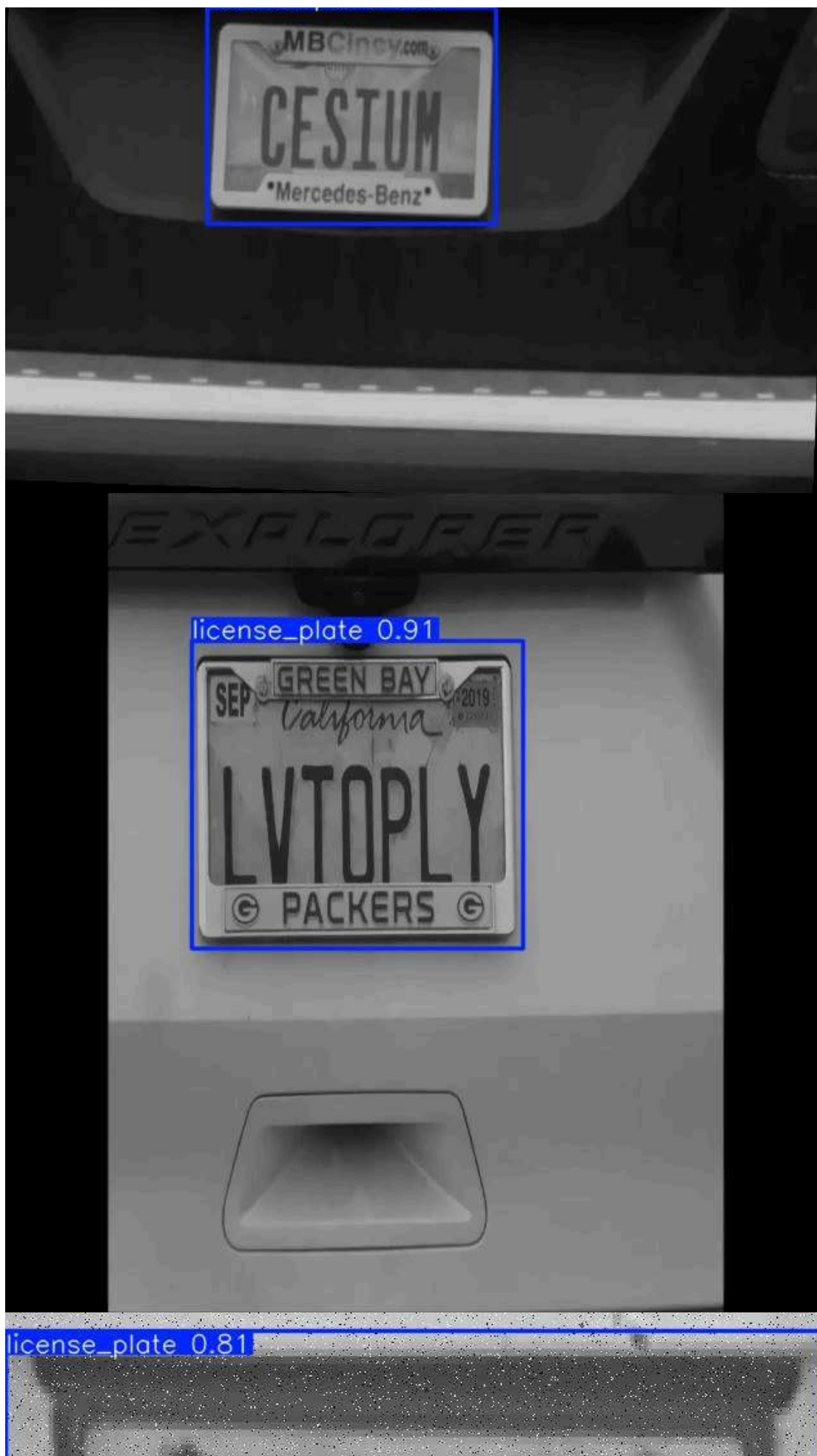








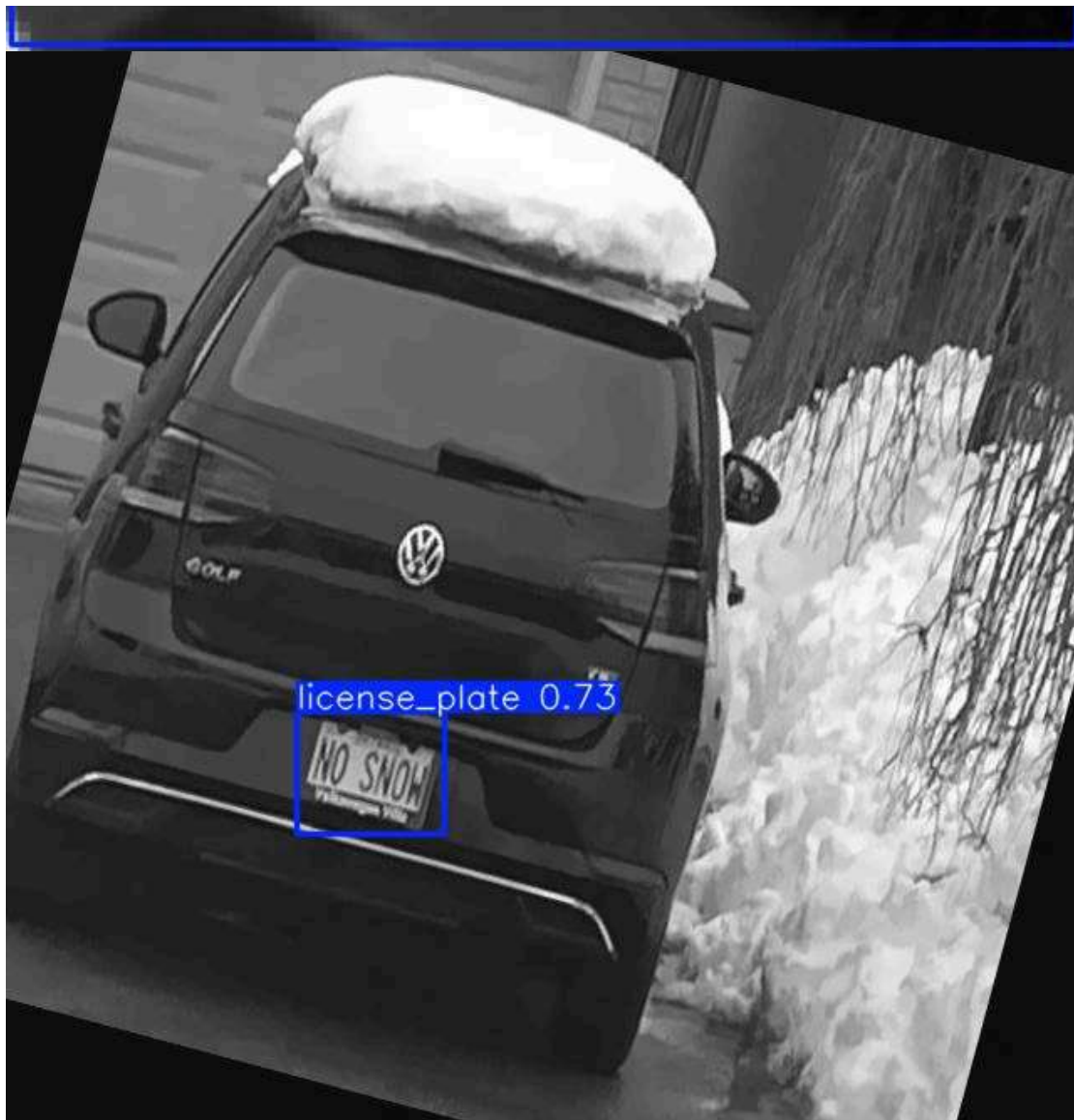




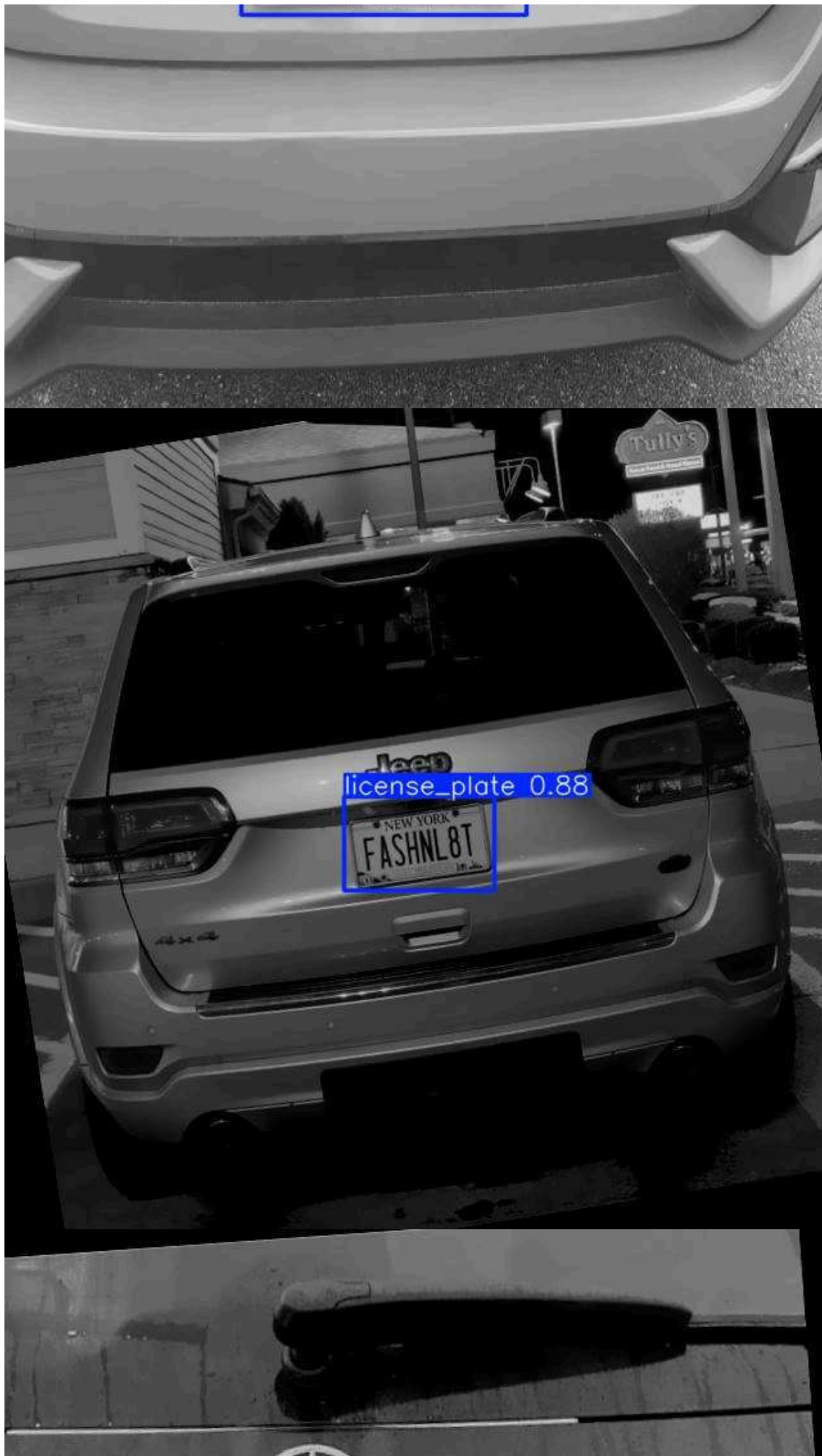


license_plate 0.90



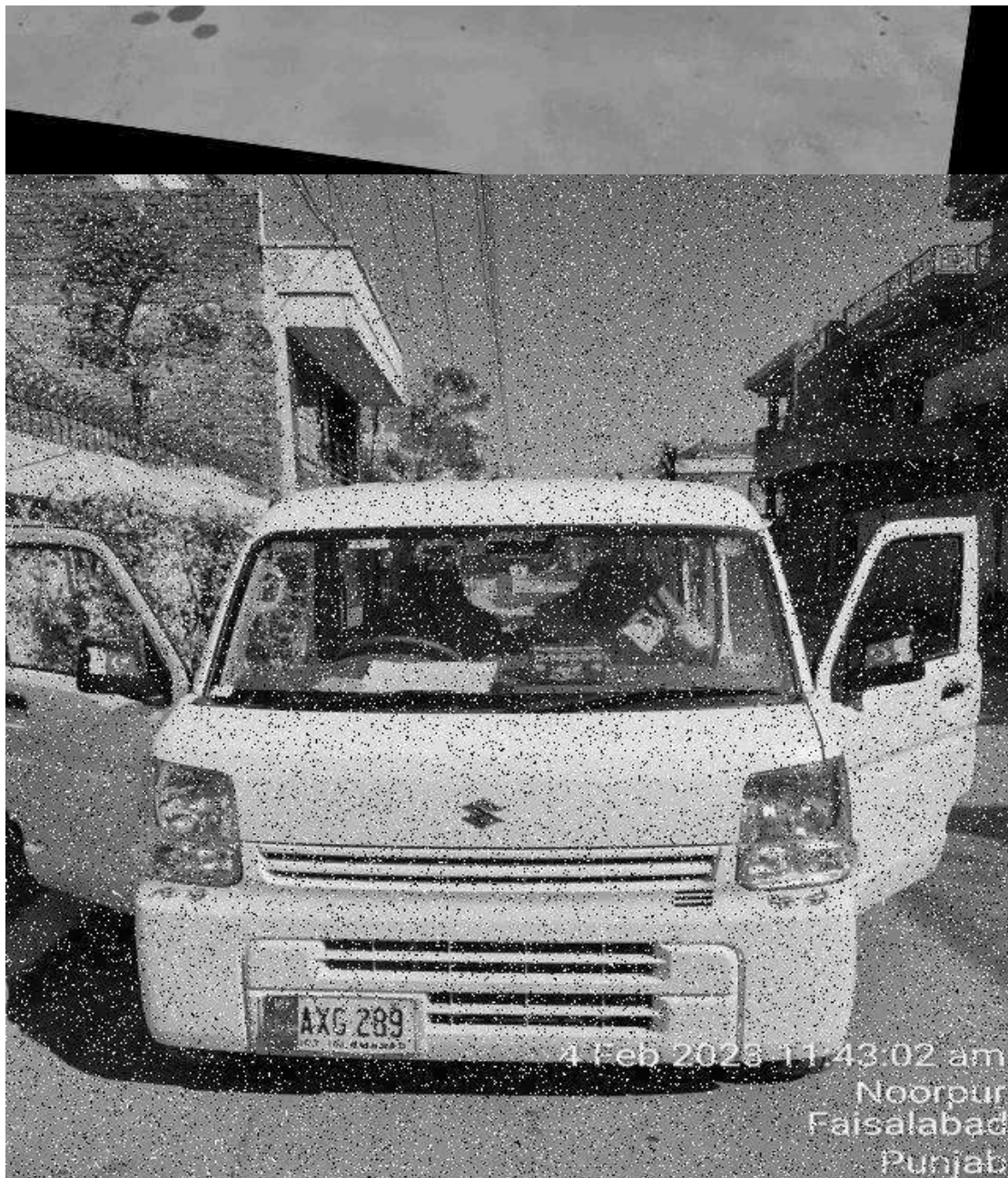




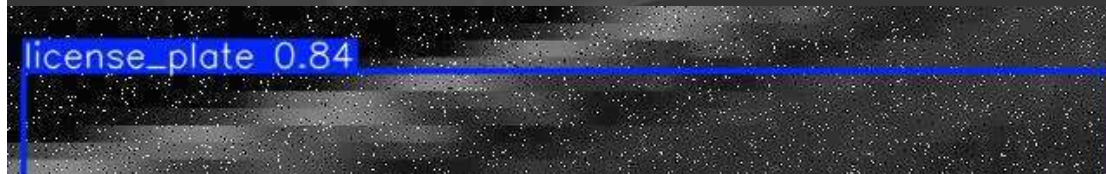














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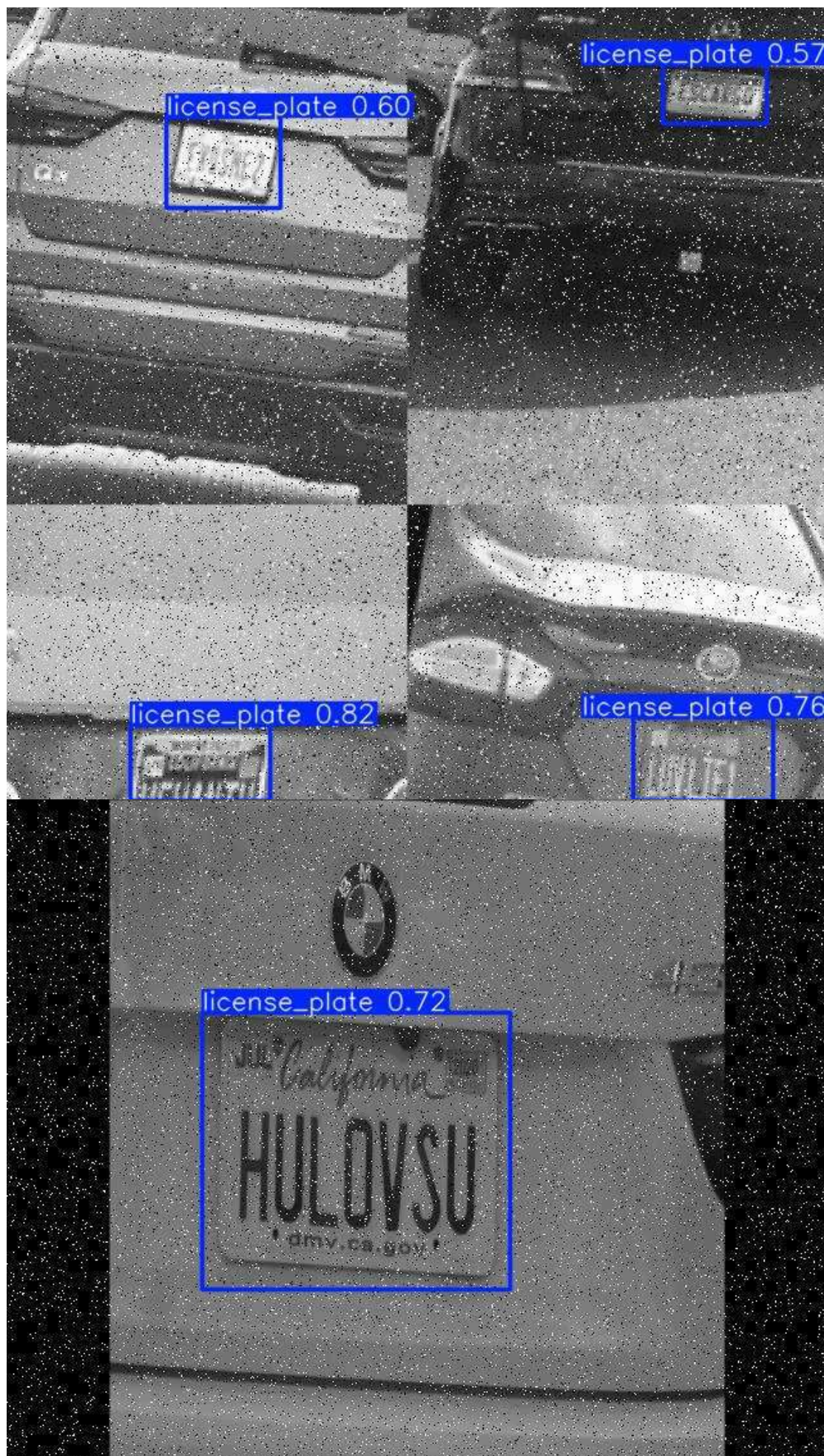


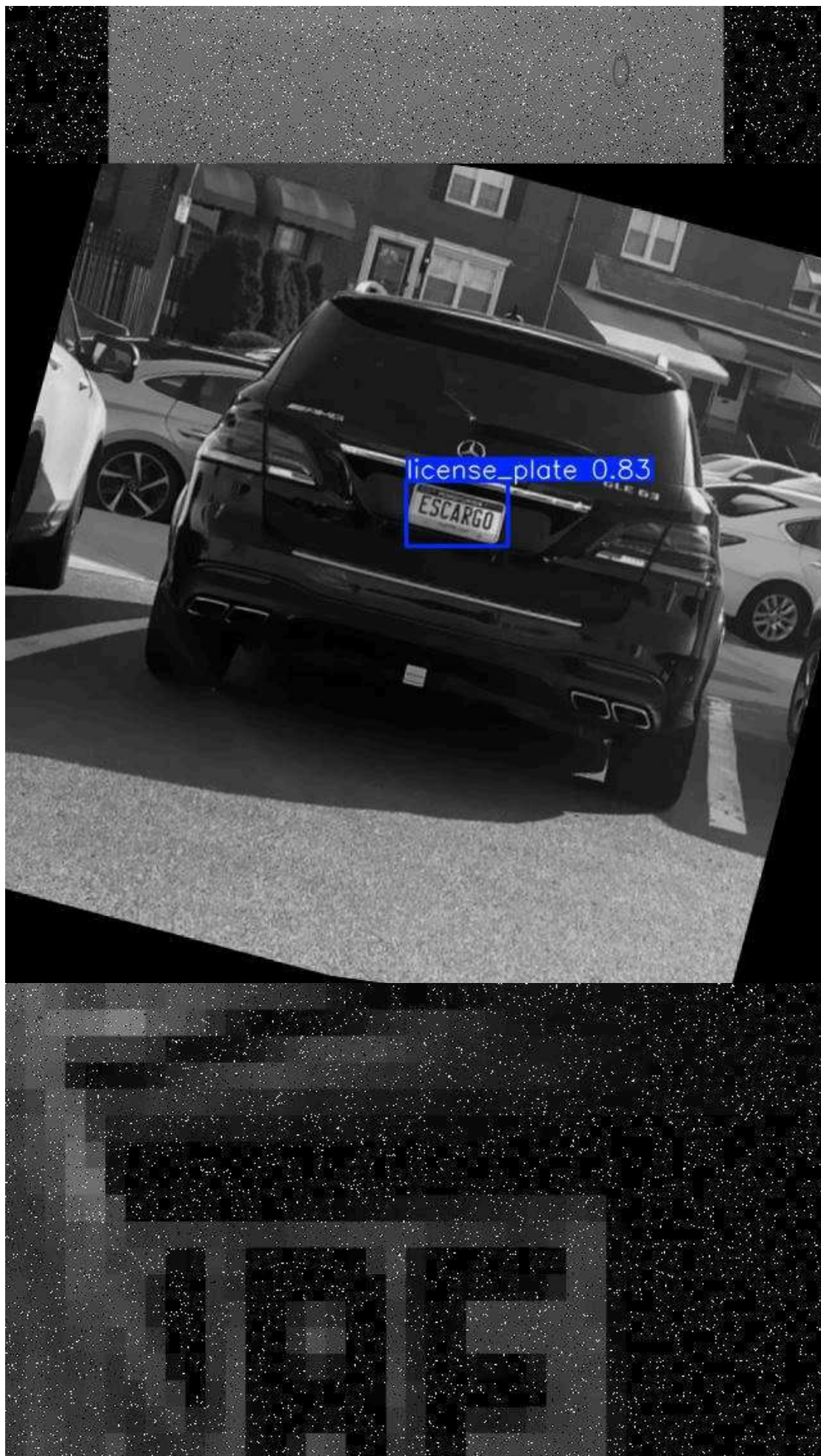
















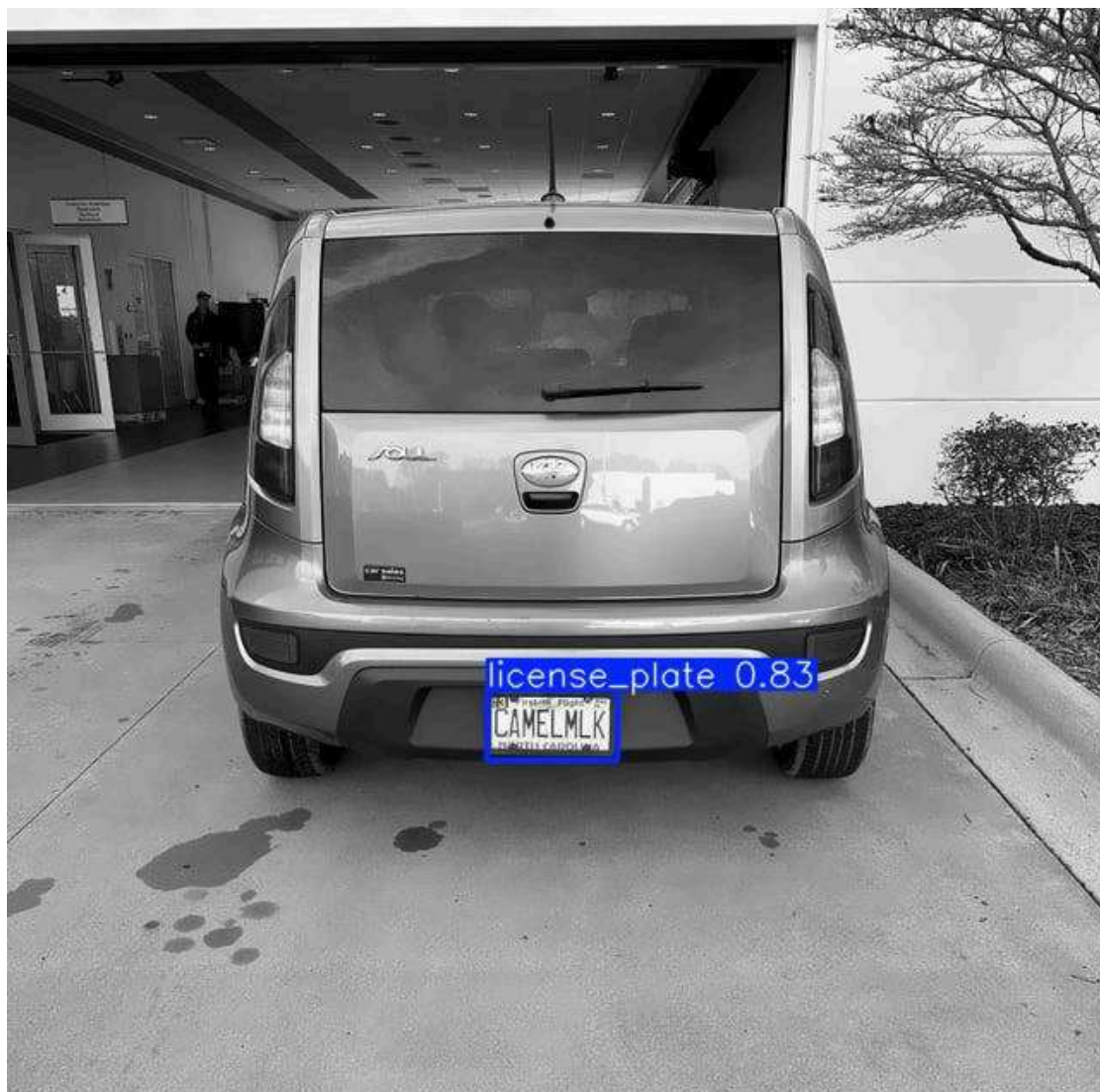


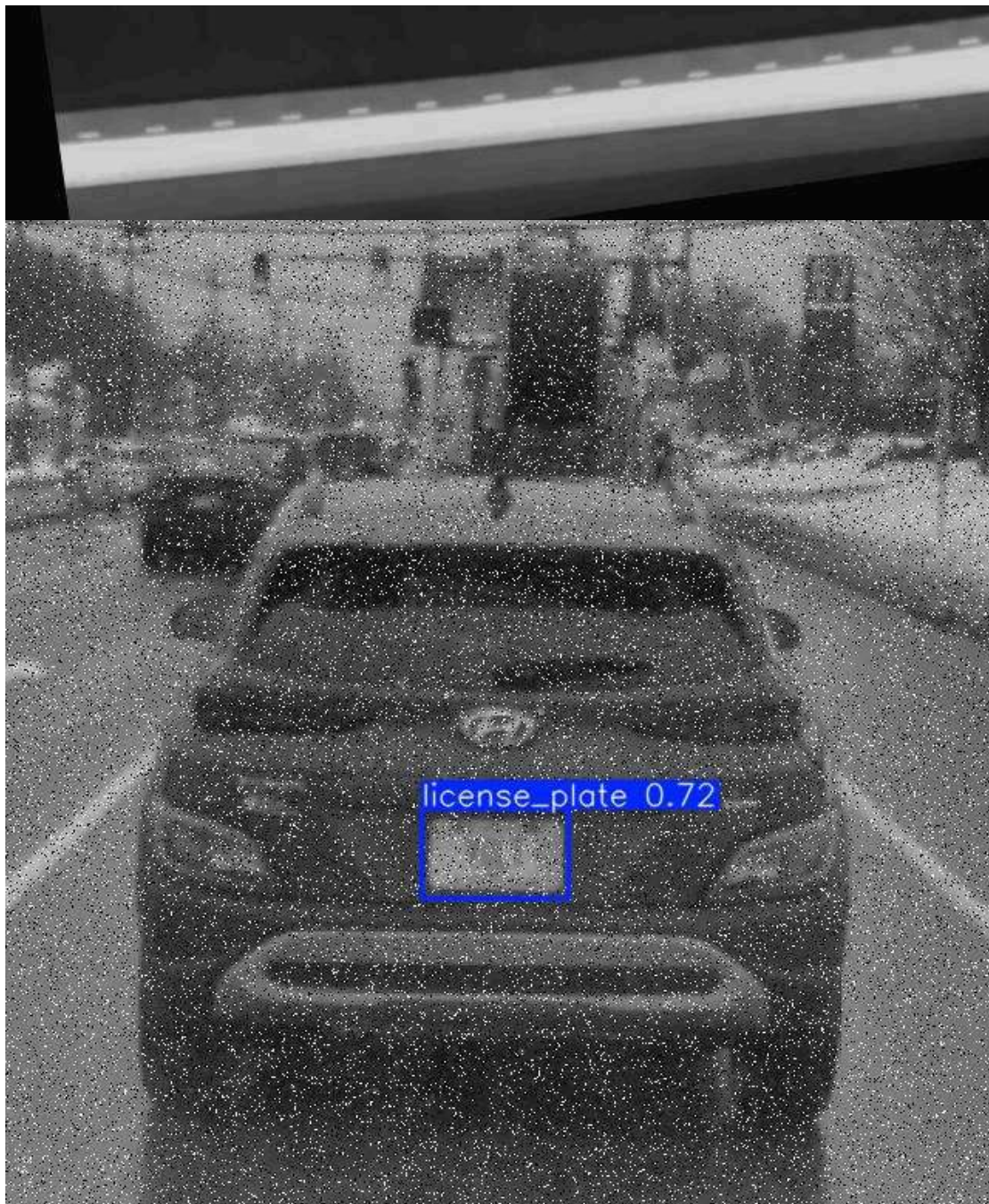


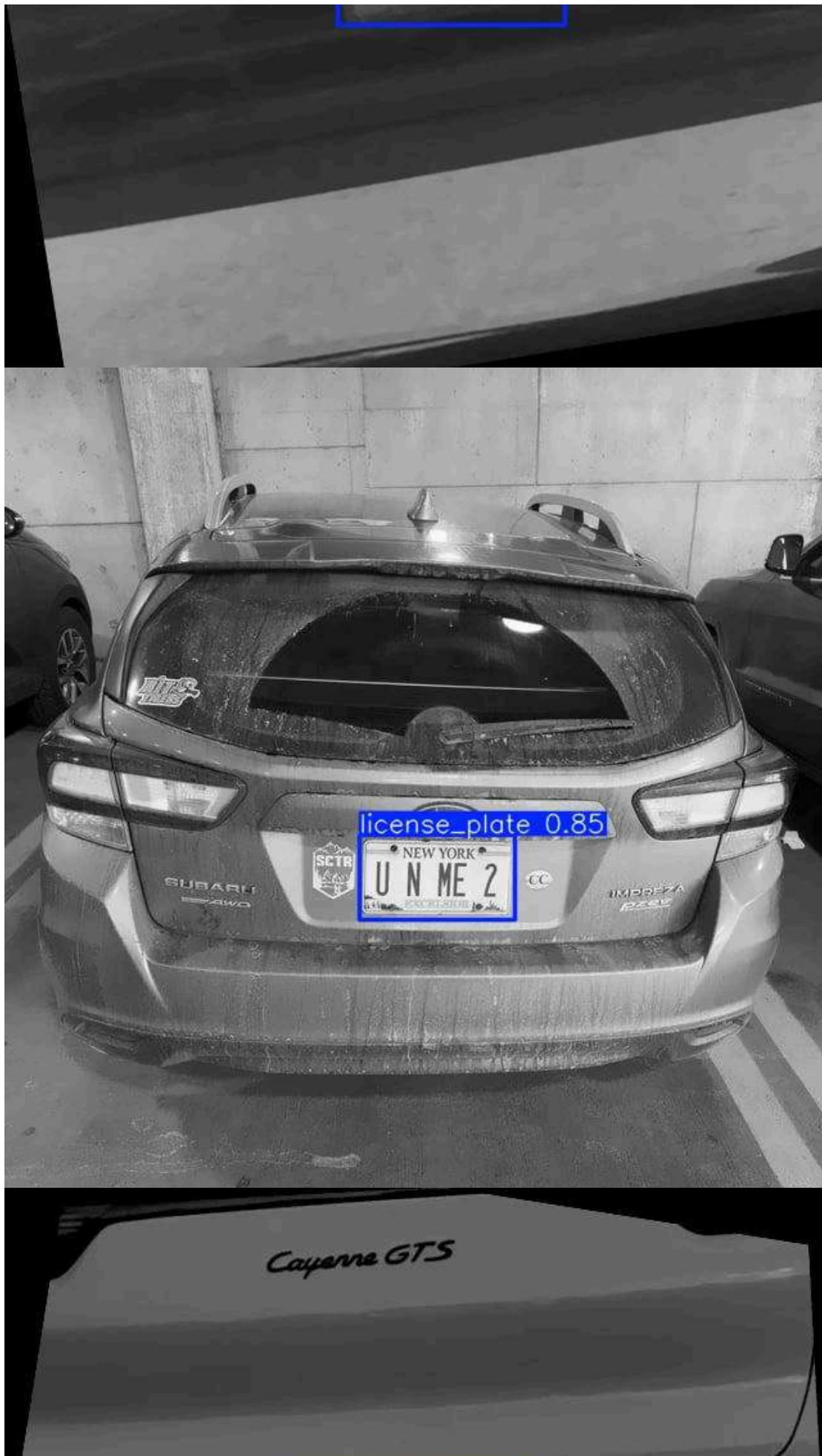




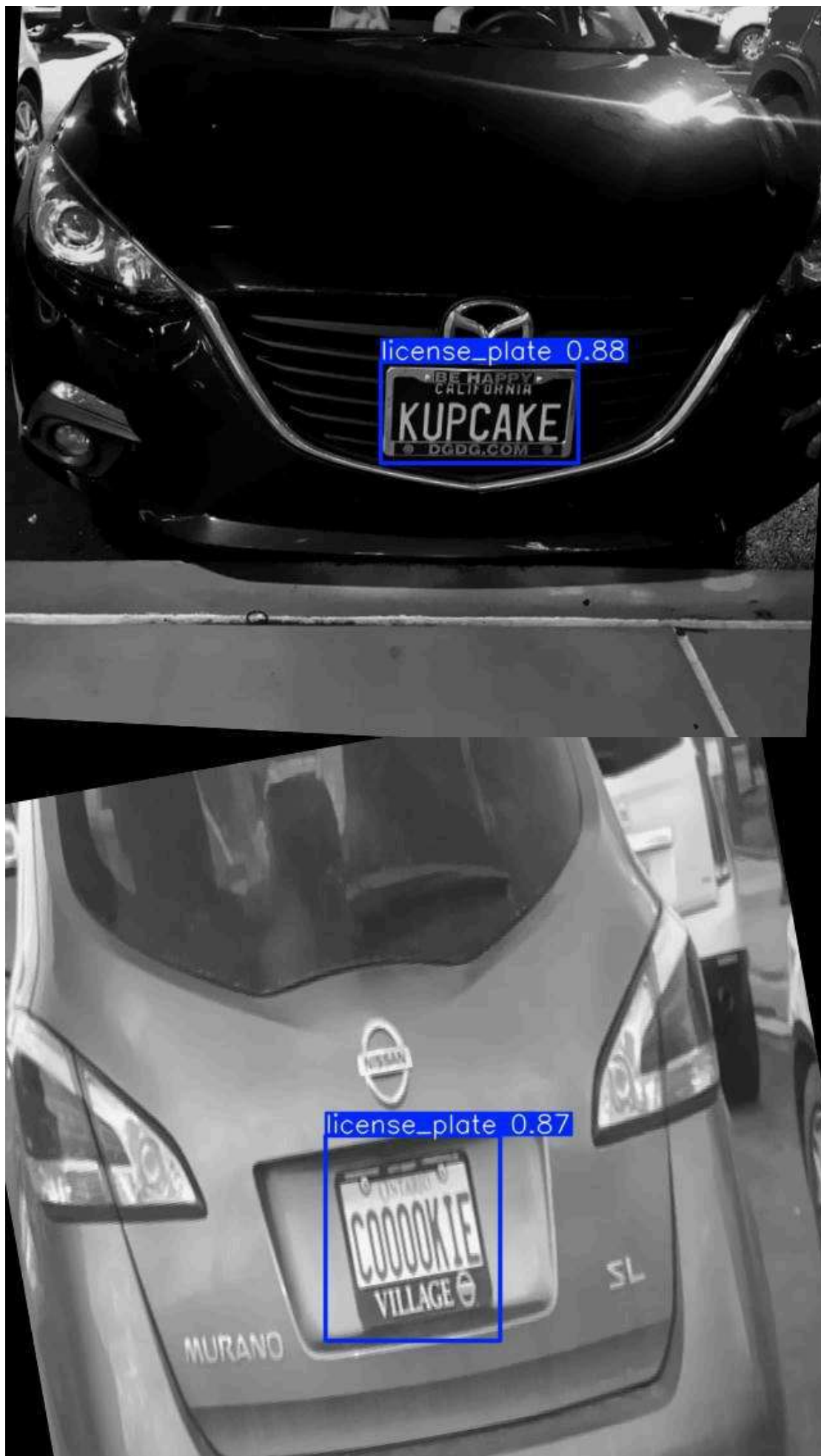




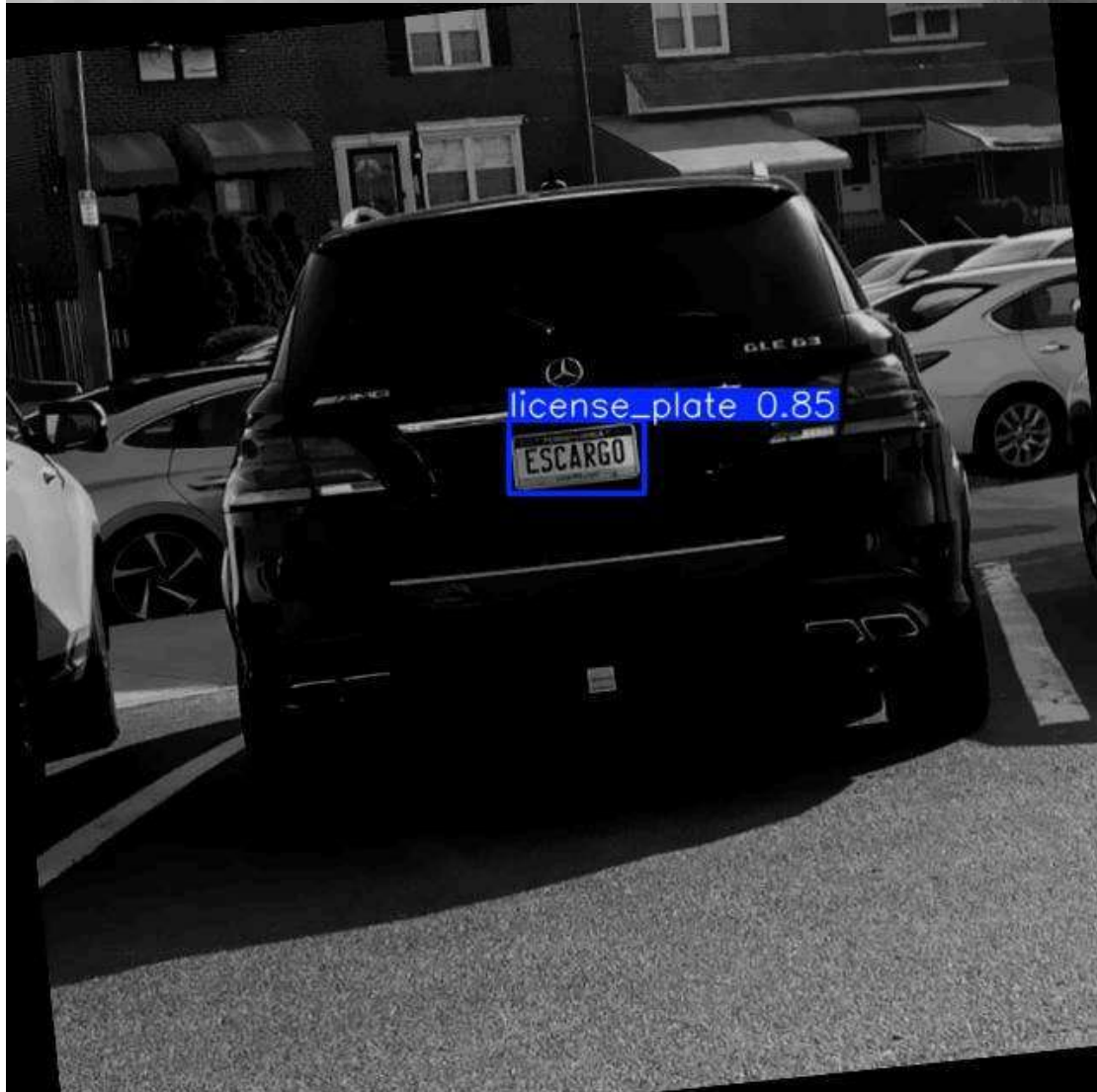










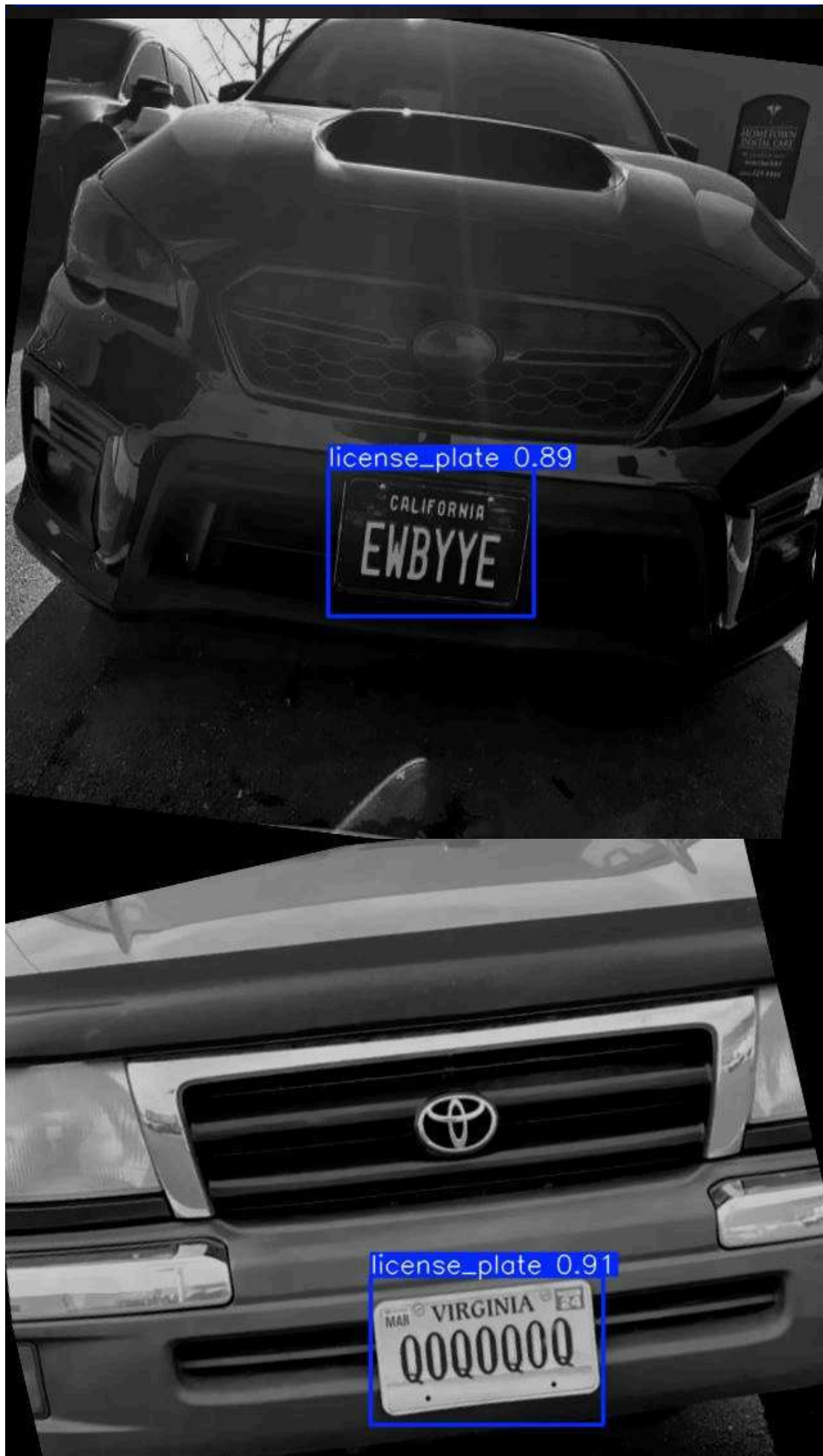




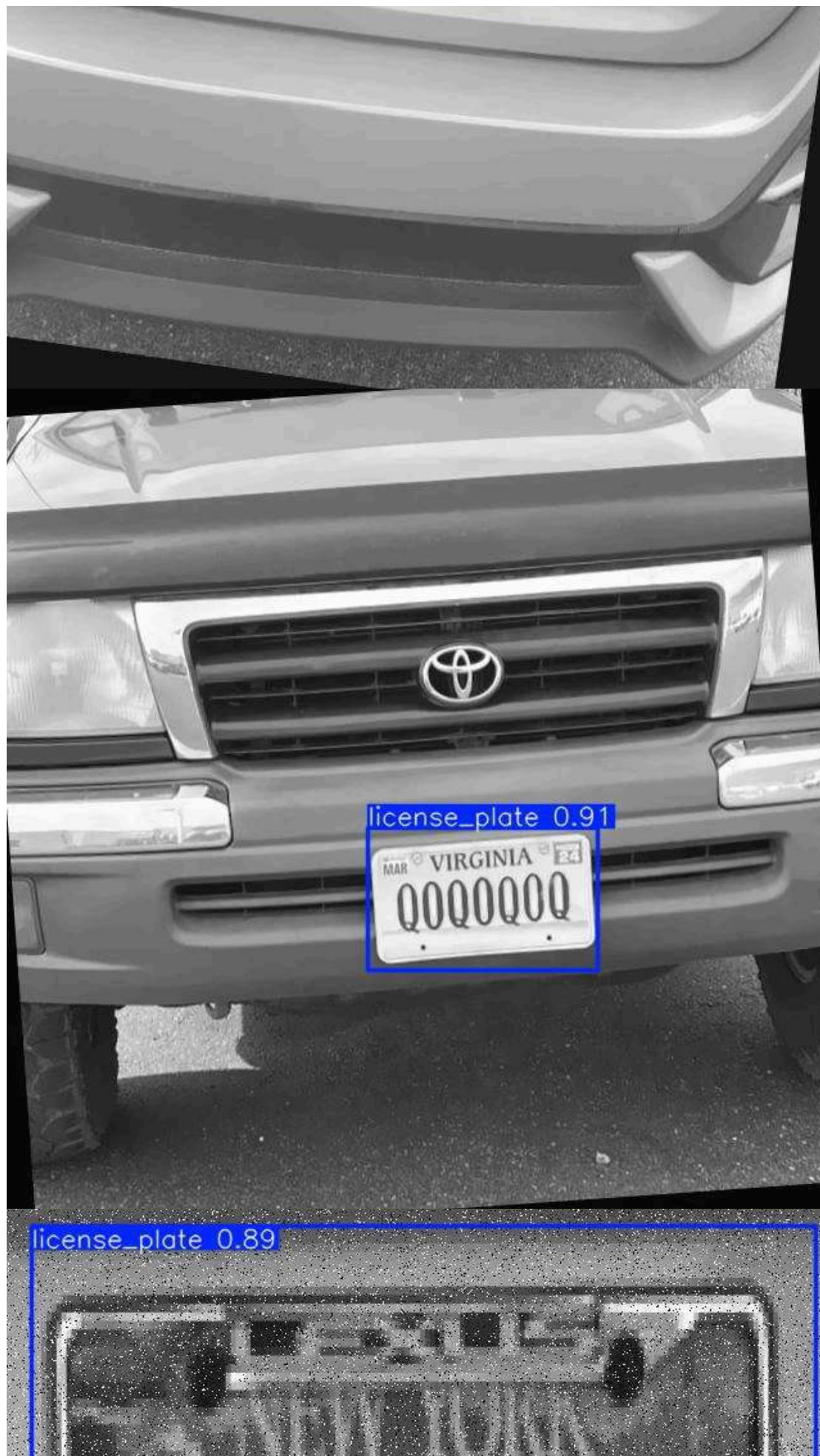


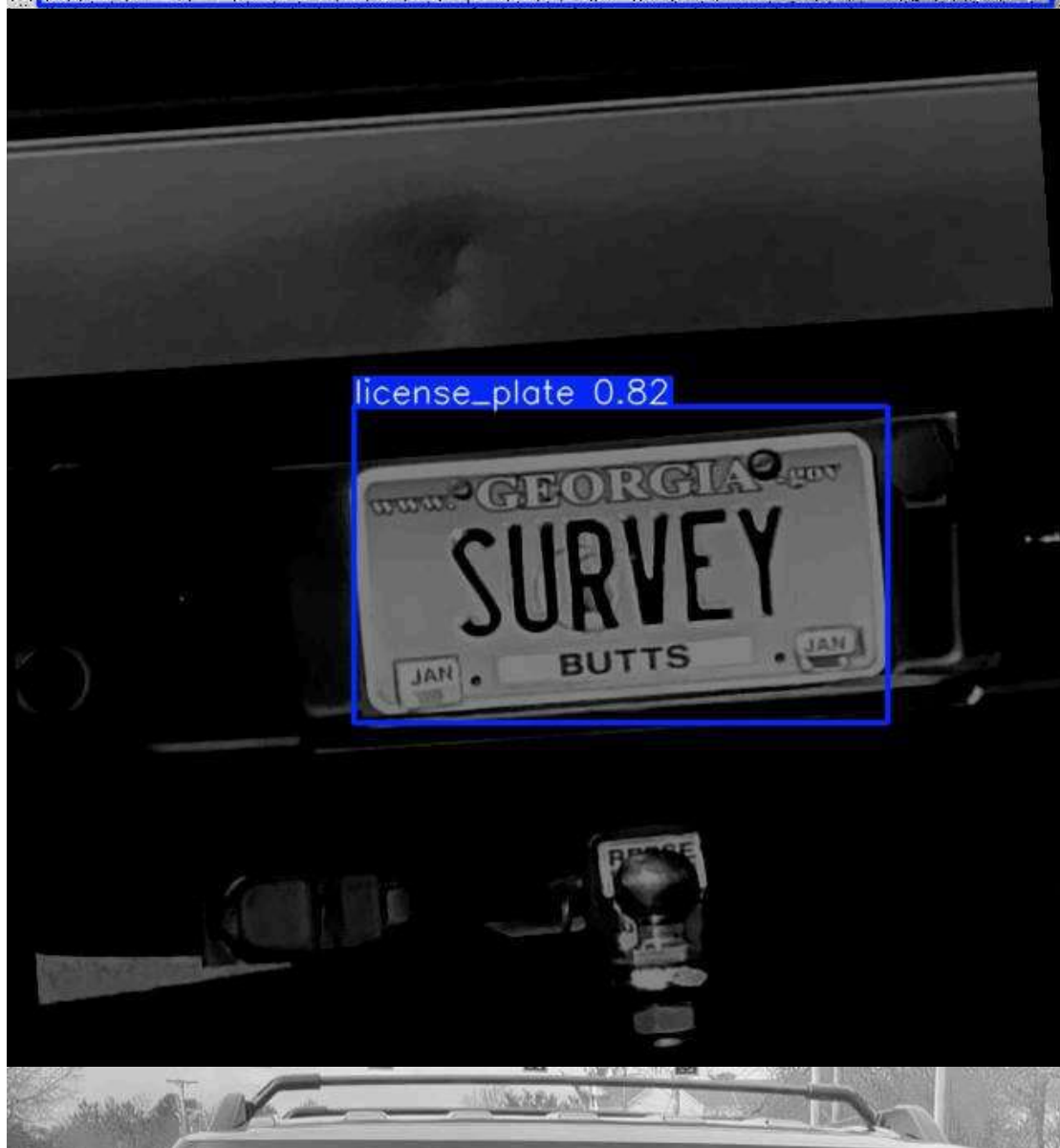
license_plate 0.88



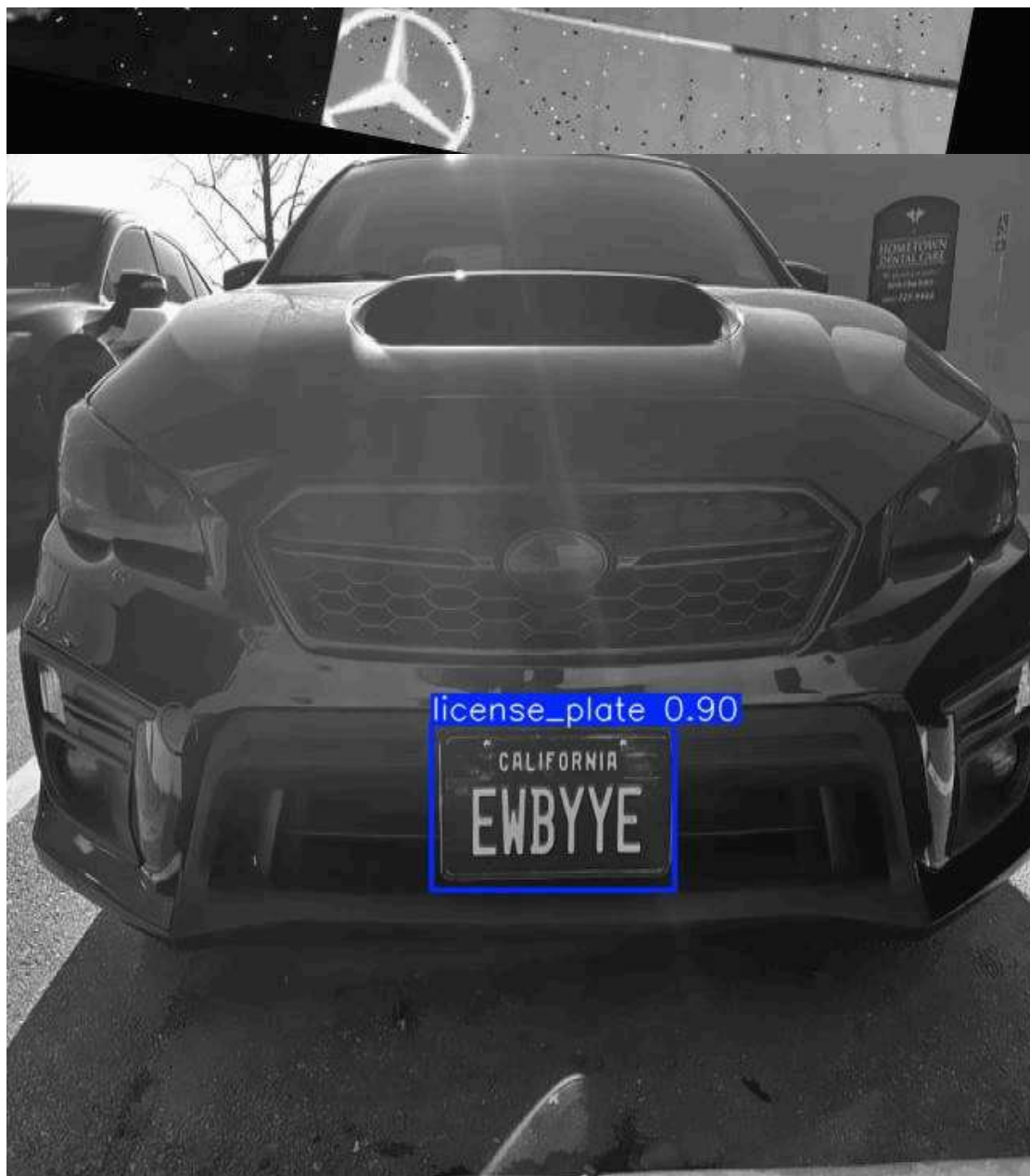








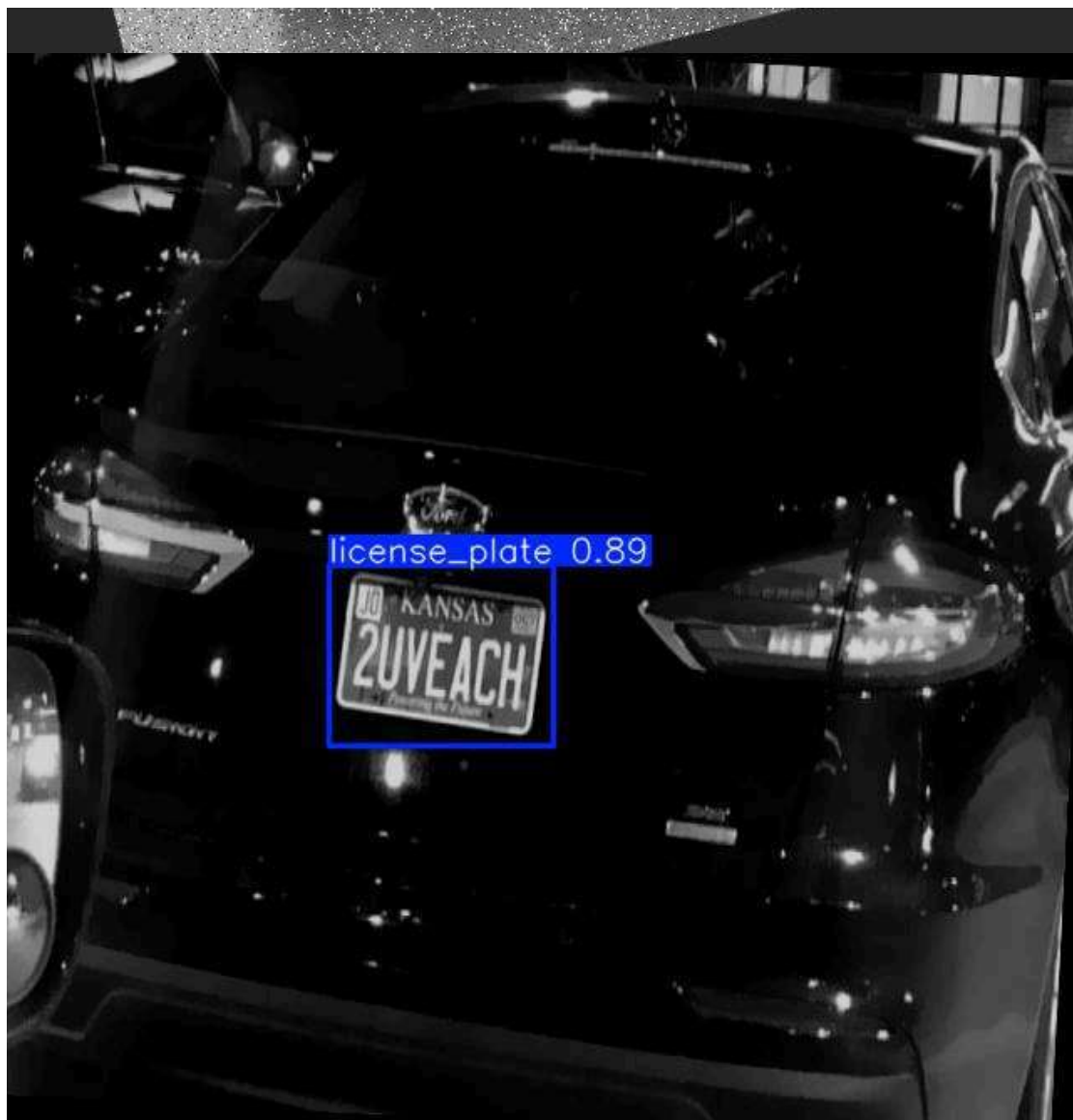


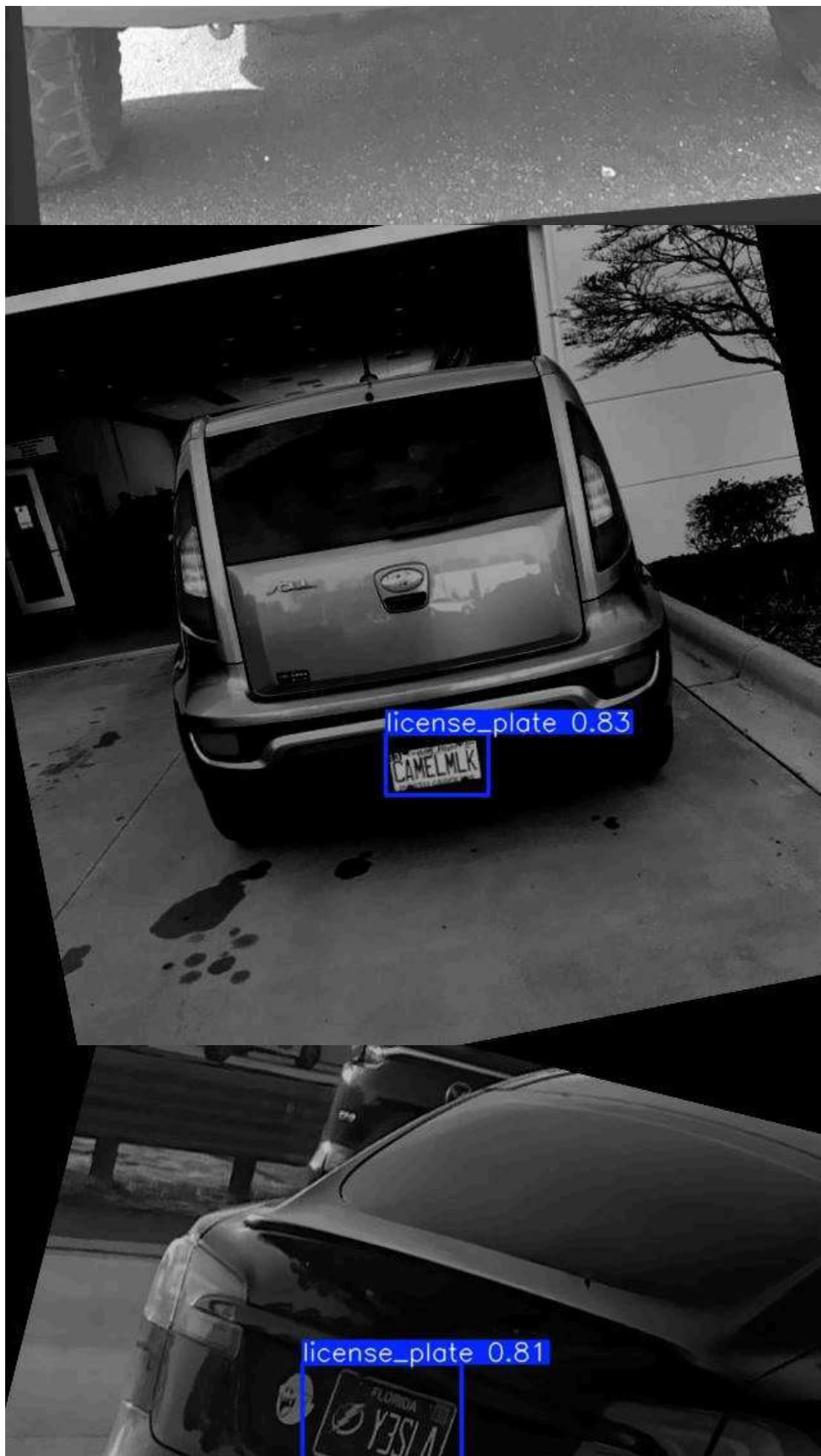


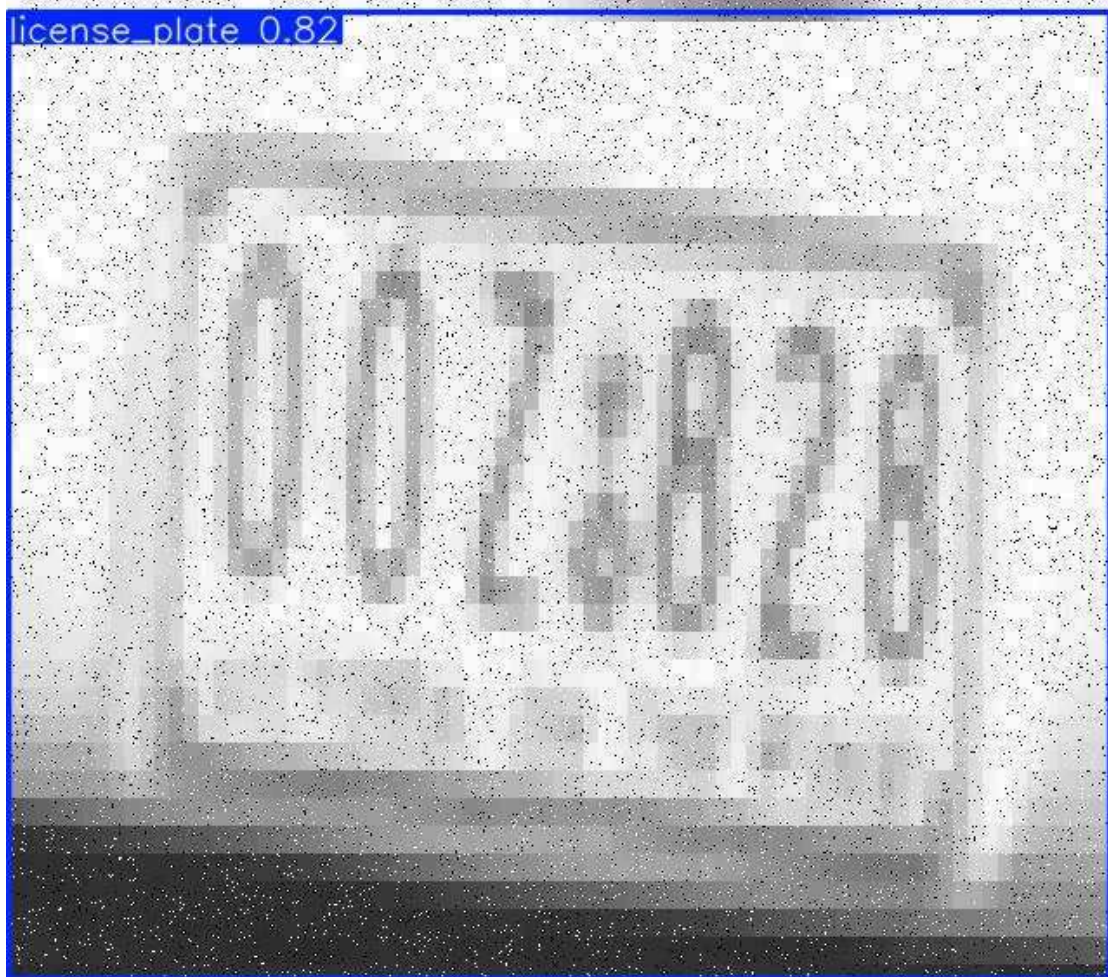


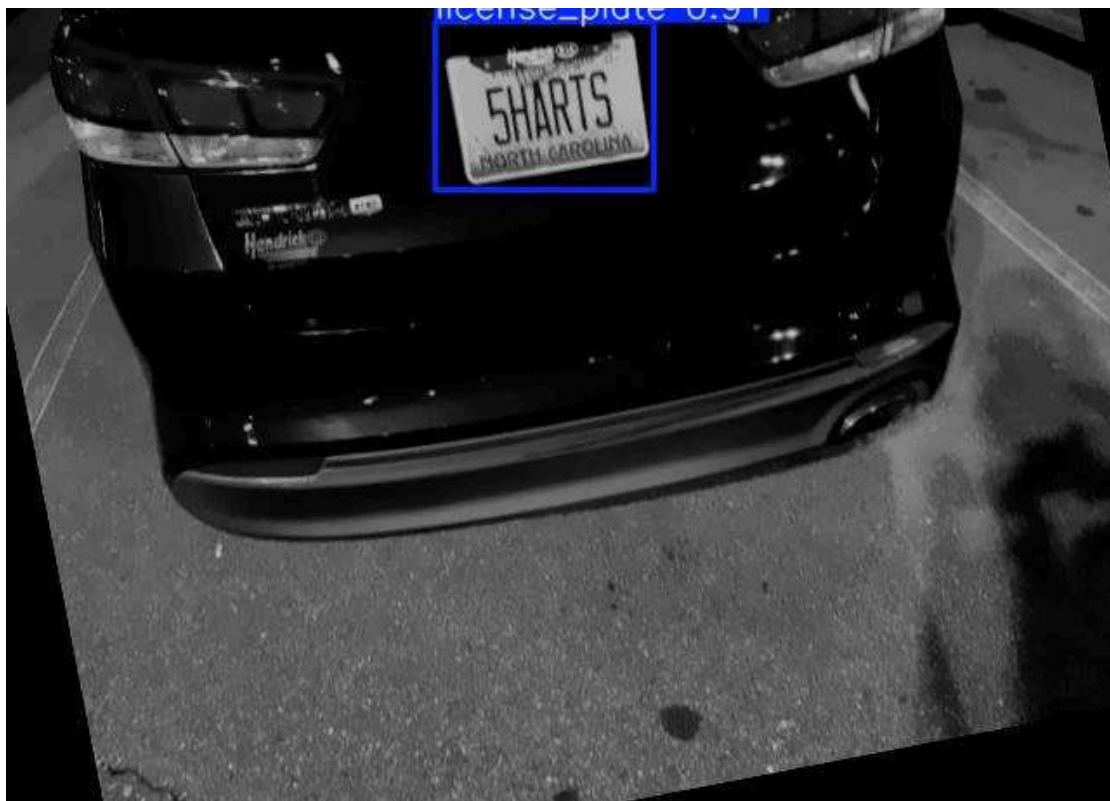


















```
from ultralytics import YOLO

# Load the trained model
model = YOLO('runs/detect/train/weights/best.pt')

# Perform validation
metrics = model.val()

# Print metrics
print("Validation Metrics:")
print(f"mAP@0.5: {metrics.box.map50:.2f}")      # Mean Average Precision at IoU 0.5
print(f"mAP@0.5:0.95: {metrics.box.map:.2f}")  # Mean Average Precision at IoU 0.5 to 0.95
print(f"Precision: {metrics.box.mp:.2f}")       # Mean Precision
print(f"Recall: {metrics.box.mr:.2f}")         # Mean Recall

# Display the saved validation results
import glob
from IPython.display import Image, display

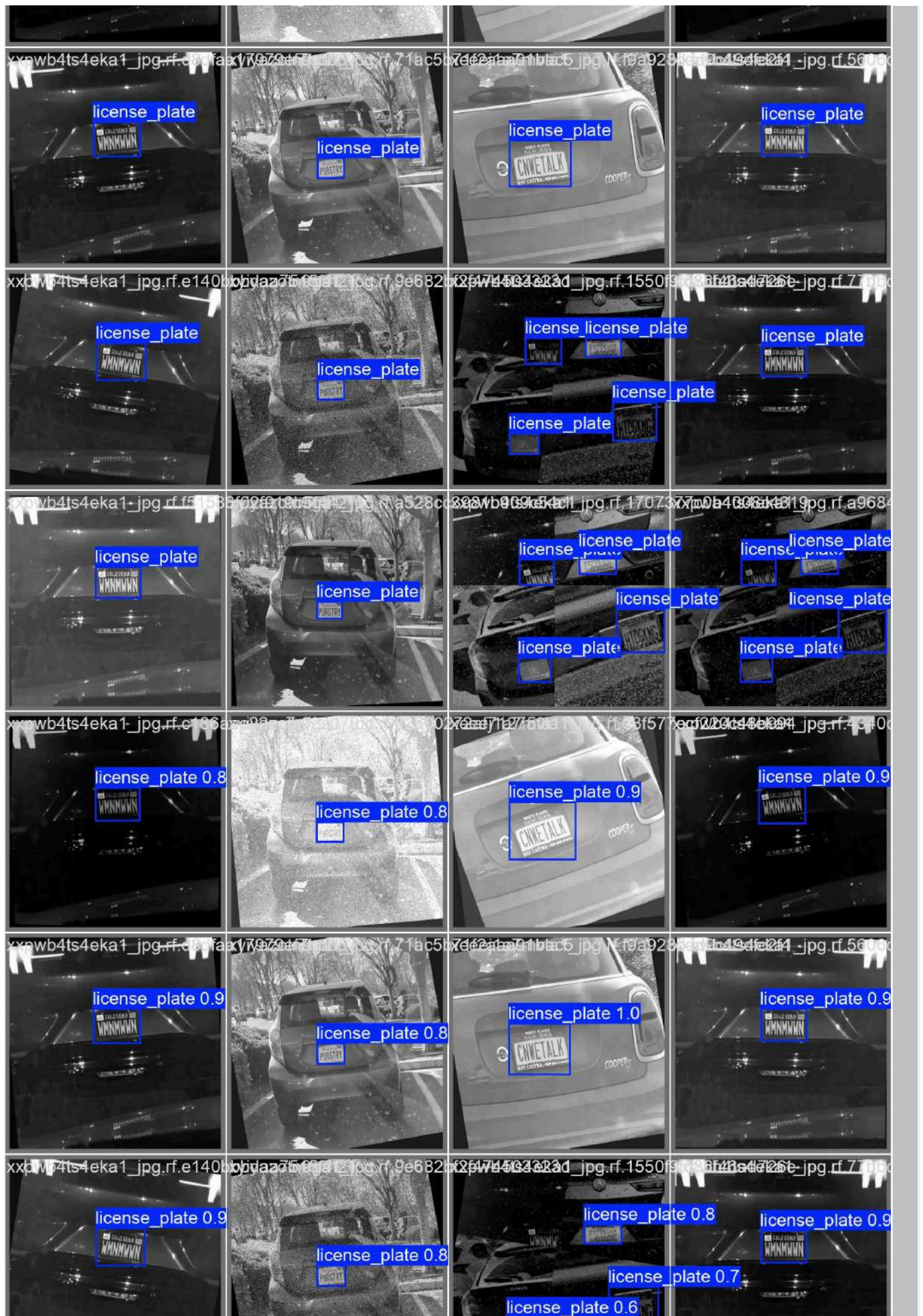
val_images = glob.glob('runs/detect/val/*.jpg') # Path where validation predictions are saved
for val_image in val_images:
    display(Image(filename=val_image))
```

Ultralytics 8.3.58 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15102MiB)
 Model summary (fused): 168 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
val: Scanning /content/drive/MyDrive/License plate detection and recognition/valid/labels.cache... 4

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	400	438	0.96	0.945	0.969	0.783

 Speed: 0.4ms preprocess, 4.1ms inference, 0.0ms loss, 2.6ms postprocess per image
 Results saved to **runs/detect/val2**
 Validation Metrics:
 mAP@0.5: 0.97
 mAP@0.5:0.95: 0.78
 Precision: 0.96
 Recall: 0.95











```
import glob
from IPython.display import Image, display

# Path to validation results
val_dir = 'runs/detect/val/'

# Display evaluation visualizations (precision-recall curves, etc.)
print("Displaying Accuracy Visualizations:")

# Show precision-recall curves
pr_curve = glob.glob(val_dir + 'PR_curve.png')
if pr_curve:
    display(Image(filename=pr_curve[0]))
else:
    print("Precision-Recall curve not found.")

# Show confusion matrix (if available)
confusion_matrix = glob.glob(val_dir + 'confusion_matrix.png')
if confusion_matrix:
    display(Image(filename=confusion_matrix[0]))
else:
    print("Confusion matrix not found.")

# Additional images in the val directory
additional_images = glob.glob(val_dir + '*.png')
for img in additional_images:
    if img not in pr_curve and img not in confusion_matrix:
        display(Image(filename=img))
```

↔ Displaying Accuracy Visualizations:

